Gravitational potential, particles and halos: GADGET-4 versus PKDGRAV3 from identical initial conditions

32^3 and 256^3 particles in $200\,h^{-1}$ Mpc, Sahyadri default settings on pawna

Sahyadri sandbox, branch `rockstar-pkc-reader`, tag `potcmp-256-report`

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Abstract

GADGET-4 and PKDGRAV3 were run from identical initial conditions (GADGET-4 NGenIC, Sahyadri default settings, $200\,h^{-1}$ Mpc) with 32^3 and 256^3 particles. Each run went through the full Sahyadri halo pipeline (Rockstar, consistent-trees, VAHC), and the two codes are compared particle by particle (potential, position, velocity), halo by halo, and through the stacked potential around halos in mass bins out to $50\,h^{-1}$ Mpc. At 256^3:

- **Particles.** At $z = 12$ the potentials agree to 6×10^{-3} rms (correlation $1 - 6 \times 10^{-6}$). The difference grows to 2% at $z = 3$ and 22% at $z = 0$, and 75–90% of it comes from the particles ending up in different places: at $z = 0$ the median separation is $0.12\,h^{-1}$ Mpc and the median velocity difference $70\,\mathrm{km\,s^{-1}}$, dominated by halo particles. Once PKDGRAV3's softened self-potential is removed, the zero points agree to $\leq 1.2 \times 10^3\,(\mathrm{km\,s^{-1}})^2$.
- **Halos.** Mass functions agree within 2–3% above $10^{12}\,h^{-1}\,\mathrm{M}_\odot$. Matched halos have the same masses, spins and VAHC bias, but PKDGRAV3 halos are 20–25% more concentrated at 10^{12}–$10^{13}\,h^{-1}\,\mathrm{M}_\odot$, with 2–5% higher $V_{\max}$ and 4–9% deeper $\phi_{\min}$. This is consistent with PKDGRAV3's smaller effective softening under the production settings. For the same particles, the mean potentials of matched halos agree to 0.5%.
- **Stacked potential.** Around halos the stacked mean potential agrees to 1–2% ($\leq 0.02\,\sigma_\phi$) from 0.2 to $50\,h^{-1}$ Mpc at $z = 3$, 1, 0.5 and 0. Inside $0.1\,h^{-1}$ Mpc PKDGRAV3 is 3–6% deeper. The volume-weighted mean (potential on a 512^3 grid) is much shallower than the particle-weighted one and tends to zero far from the halos. Its code difference is within 2 paired errors ($\leq 0.016\,\sigma_\phi$) at every radius from 0.4 to $50\,h^{-1}$ Mpc, in both 36 log bins and 250 linear bins of $0.2\,h^{-1}$ Mpc.
- **Pipeline per-halo potential (`.vpot`).** VAHC was rerun for both codes with `vahc_fast.py`. It now also writes each halo's mean particle potential within r_s, R_{200b}, 0.5–1.5 R_{vir} and at the seven VAHC scales, for GADGET-4 too (from its HDF5 `Potential`). For matched halos at $z = 0$, PKDGRAV3 is deeper by a median 3–6% within r_s, 1.3–2% within R_{vir} and ∼1% at $5\,h^{-1}$ Mpc (above $10^{12}\,h^{-1}\,\mathrm{M}_\odot$). The outer ∼1% is the mass-weighted mean of the deeper halo cores, not a zero-point offset.
- **Halo–halo potential cross-correlation.** A new tool measures, between two mass bins, the mean difference, structure function and correlation of the per-halo potentials as a function of separation. It is C with OpenMP and dilution, built for 2048^3, and is an optional pipeline stage. At 256^3 the two codes agree within the jackknife errors at all separations from 0.4 to $50\,h^{-1}$ Mpc. Within the lowest mass bin the correlation falls from 0.999 at $0.4\,h^{-1}$ Mpc to 0.28 at $50\,h^{-1}$ Mpc.
- **Correlation multipoles.** The halo cross-correlation multipoles ($\ell = 0$–4, dipole included) between mass bins, from 0 to $100\,h^{-1}$ Mpc, were measured in real space, in redshift space, and with the gravitational-redshift shift of each halo's potential. They use Corrfunc with signed μ (the `arematti/Corrfunc@desi` fork) and are a pipeline stage. The codes agree within errors, and redshift space shows the Kaiser monopole boost and a negative quadrupole. The potential adds a small positive dipole between bins of different mass, measured as a difference of spaces. It is consistent between the codes and linear in the potential (checked with the shift ×50).

The stacking tool keeps only per-mass-bin sums and dilutes the outer shells by particle ID. A stack of a 256^3 snapshot with $\sim 17\,000$ halos takes $\sim 10\,\mathrm{s}$; gridding the potential for the volume weighting takes 6 min.

Contents

1 Set-up **2**

2 Where things are **3**

3 Methods **5**

4 Results at 32^3 **8**

5 Results at 256^3 **10**

5.1 Particles: potential, positions, velocities 10

5.2 Halo catalogues and VAHC 11

5.3 Stacked potential around halos 13

5.4 Volume-weighted stacks and linear bins 15

5.5 Per-halo potential from the pipeline (`.vpot`) 17

5.6 Halo–halo potential cross-correlation 18

5.7 Correlation multipoles in real and redshift space 20

6 Problems found on the way **22**

7 Reproducing **22**

1 Set-up

Both codes evolve the *same* particle load. GADGET-4's NGenIC (2LPT, CLASS $P(k)$ at $z = 49$, the Sahyadri production recipe of `scripts/submit_CiPod.sh`) writes the initial conditions; the same file, converted to GADGET-2 format, is read by PKDGRAV3 (`achInFile`), so the two runs differ only in the N-body solver, the time stepping and the softening. Every setting is the Sahyadri default of `config/submit/sahyadri/sub_default.conf` unless stated (Table 1).

box, particles	$L = 200\,h^{-1}$ Mpc; $N = 32^3$ (test) and 256^3
cosmology	$\Omega_m = 0.3137721$, $\Omega_b = 0.0493017$ (GADGET-4 parameter only; both runs are dark matter only), $\Omega_\Lambda = 0.6862279$, $h = 0.6736$, $n_s = 0.9649$, $A_s = 2.098903 \times 10^{-9}$, seed 12345678
initial conditions	GADGET-4 NGenIC 2LPT at $z = 49$ (CLASS spectrum, no renormalisation)
outputs	Sahyadri's 100 output times (`config/sims/outputs100.txt`, $z = 12 \to 0.0093$) plus $a = 1$: 101
GADGET-4	commit 1e171a4, production Config + `EVALPOTENTIAL`; TreePM, PMGRID $= 2N^{1/3}$, `ErrTolForceAcc` 0.002, $\theta_{\max} = 0.75$; softening $\epsilon_G = L/(30N^{1/3})$ (Plummer-equivalent)
PKDGRAV3	`shadaba/pkdgrav3` `main` (binary built from 5766bb28; later commits are documentation); production `sahyadri2048.par` settings: $\epsilon_P = 1/(60N^{1/3})$ box units, 20 steps between outputs, output at the redshifts GADGET-4 actually wrote; `dCompPotStep=1` (potential in the `.pkc` files)
softening	256^3: $\epsilon_G = 26.0\,h^{-1}$kpc, $\epsilon_P = 13.0\,h^{-1}$kpc ($32^3$: 208 and $104\,h^{-1}$kpc); PKDGRAV3's force is Newtonian beyond $2\epsilon_P = \epsilon_G$, GADGET-4's (spline) beyond $2.8\,\epsilon_G$
halos	Rockstar (production template: 30 particles output, 20 minimum, unbound threshold 0.5; GADGET-4 HDF5 via `FILE_FORMAT = "AREPO"`, PKDGRAV3 via the native `.pkc` reader), consistent-trees, VAHC (grid 512^3)

Table 1: Simulation and pipeline settings.

Pipeline. One command submits everything as a PBS chain (`scripts/gadget_pkdgrav3_compare/submit_compare.sh compare256.conf`): GADGET-4 job (CLASS → NGenIC → IC conversion → run) → PKDGRAV3 `sim` → Rockstar → consistent-trees → VAHC for each code (the Sahyadri PKDGRAV3 pipeline, `scripts/pkdgrav3_pipeline`) → comparison job (`job_compare.pbs`). At 32^3 consistent-trees cannot run (at most 15 halos per snapshot; it needs enough halos to estimate its consistency tolerances), so the 32^3 halo results use the Rockstar catalogues.

2 Where things are

Data on pawna

Work directory `/storage/shadab/pkdgrav3_io/potcmp/` (below: `data/` = `/storage/shadab/pkdgrav3_io/potcmp/data/`).

what	path
GADGET-4 256^3 run	`data/sims/potcmp/gadget256/r1/`: `snapshot_000..100.hdf5` (with Potential), `param_run.txt`, `gadget_run.log`, `snapnames.txt`, `gadget_out_z.txt`
initial conditions 256^3	`data/sims/potcmp/gadget256/r1/ics/`: `snapshot_ics_000.hdf5` (GADGET-4), `ics_gadget2.dat` (PKDGRAV3 input), `param_ics.txt`; CLASS: `data/transfer/class_gadget256*`
PKDGRAV3 256^3 run	`data/sims/potcmp/pkd256/r1/`: `p.par`, `snap.<step>` (HDF5), `snap.<step>.s<pct>.0.pkc`, `snapnames.txt`, `run.log`
halos 256^3	`data/halos/potcmp/gadget256/r1/`, `data/halos/potcmp/pkd256/r1/`: `out_<snap>.list` (Rockstar), `out_<snap>.trees` (consistent-trees hlists), `out_<snap>.vahc`, `trees/`, `hlists/`; `../scales.txt`
comparison 256^3	`data/compare/pkd256_vs_gadget256_r1/`: `particles.csv`, `halos.csv` (potentials of matched halos), `summary.json`, `snap_*.npz` (200 000-particle subsamples), `profiles/{gadget,pkd,pkc}_out<i>.npz` (stacks, ID-hash dilution; the job's first version with per-chunk random dilution in `profiles_jobrun_target4096/`), `halos/` (`halo_mf.csv`, `halo_match.csv`, `halo_match_out<i>.npz`), `compare.log`
32^3 (same layout)	`data/sims/potcmp/{gadget32,pkd32}/r1/`, `data/halos/potcmp/{gadget32,pkd32}/r1/`, `data/compare/pkd32_vs_gadget32_r1/` (stacks in `profiles/`; the earlier $r/R_{\rm vir}$ profiles in `profiles_rvir_old/`)
first 32^3 run (no `EVALPOTENTIAL`)	`/storage/shadab/pkdgrav3_io/potcmp/data_run1_zeropot/` (Potential $= 0$; kept as evidence)
job logs	`data/logs/<stage>.potcmp_{gadget,pkd}256_r1.<job>.log`, chains `data/logs/chain.*.txt`
GADGET-4 builds	`/storage/shadab/pkdgrav3_io/potcmp/g4home/code/Gadget-4/{mesh512, mesh512-NGenIC,mesh64,mesh64-NGenIC}/Gadget4`
PKDGRAV3 binary	`/storage/shadab/pkdgrav3_io/build/pkdgrav3`

Scripts and report (repository `a-paranjape/sahyadri-sandbox`, branch `rockstar-pkc-reader`; tag `potcmp-256-report`)

`scripts/gadget_pkdgrav3_compare/`	
`compare32.conf`, `compare256.conf`, `*_gadget.conf`	run configurations (one per resolution; the `_gadget` file switches to the GADGET-4 side)
`submit_compare.sh`	submits the whole chain
`job_gadget.pbs`, `setup_gadget4_pawna.sh`, `Config.sh.template`, `make_gadget4_param.py`	GADGET-4 build, ICs and run, Sahyadri recipe
`gadget4_ic_to_gadget2.py`	the GADGET-4 ICs for PKDGRAV3 (IDs → iOrder, mass so that PKDGRAV3's total is Ω_m)
`job_compare.pbs`	comparison job: the three scripts below
`compare_potential.py`	particle by particle: potential, positions, velocities; FFT Poisson reference
`halo_potential_profile.py`	stacked potential around halos in mass bins (Section 3)
`compare_halos.py`	mass functions, matched halos, VAHC quantities
`compare_vpot.py`	per-halo pipeline potential (`.vpot` columns of the VAHC FITS files) of matched halos
`scripts/post-process/vpot_hdf5.py`	`.vpot` for GADGET-4 HDF5 snapshots (used by `run_addvalue.py--fast--pot`)
`scripts/post-process/halo_potential_xcorr.py`, `halo_pair_sums.c`	halo–halo potential cross-correlation between mass bins (Section 3); pipeline stage `xcorr` (`job_xcorr.pbs`)
`scripts/post-process/halo_xi_multipoles.py`	halo cross-correlation multipoles in real and redshift space, with the gravitational-redshift shift (Section 3); stage `ximult` (`job_ximult.pbs`)
`reports/gadget_pkdgrav3_potential/`	this report: `report.tex`, `stack_potential.ipynb` (stacked potential figures), `make_figures.py` (other figures), `make_vpot_figures.py` (Section 5.5), `make_xcorr_figures.py` (Section 5.6), `make_ximult_figures.py` (Section 5.7), `figs/`, `data/` (copies of the small comparison outputs: `particles.csv`, `profiles/`, `halos/*.csv`)

3 Methods

Units and zero point of the potential. All potentials are peculiar and physical, in $(\mathrm{km\,s^{-1}})^2$. GADGET-4: ϕ = `Potential`$/a$ (attribute `a_scaling` $= -1$; the self-interaction is excluded and the periodic potential has volume mean zero). PKDGRAV3: ϕ = `Potential`$\,U^2/a$ with $U = 100\,L/\sqrt{8\pi/3}\,\mathrm{km\,s^{-1}}$ ($G = 1$, $\rho_{\rm crit} = 3/8\pi$, $L = 1$); PKDGRAV3 includes each particle's softened self-interaction $-\frac{105}{64}\,m/\epsilon_P$, a constant for equal masses, which is removed. The two codes can still differ by a constant per snapshot (zero point of the periodic sum, e.g. the Madelung-like term of a particle's own images). It changes no force, so it is measured as the median of $\phi_{\rm pkd} - \phi_{\rm Gadget}$ over all particles and removed ("aligned"). The `.pkc` potential is PKDGRAV3's, quantised to a step of $1\,(\mathrm{km\,s^{-1}})^2$ (maximum error 0.5, checked at every output).

Velocities. Peculiar, $\mathrm{km\,s^{-1}}$: GADGET-4 `Velocities`$\sqrt{a}$, PKDGRAV3 `Velocities`$\,Ua$ (at $z = 12$ the two rms velocities agree to 0.4%).

Particles are matched by ID (GADGET-4 ID -1 = PKDGRAV3 iOrder). For every 5th output (every output at 32^3) `compare_potential.py` reports the correlation and rms difference of ϕ, $|\Delta\vec{x}|$ and $|\Delta\vec{v}|$ percentiles, and, as an independent reference, the potential from an FFT Poisson solution of each snapshot's own particles (CIC on $\min(4N^{1/3}, 512)^3$, deconvolved). The rms difference between the FFT potentials of the two codes' particle positions isolates the part of $\Delta\phi$ that comes from the particles being in different places rather than from the force solver.

Stacked potential profiles. `halo_potential_profile.py` computes, for host halos (`pid` $= -1$) in bins of $M_{\rm vir}$, the mean $\langle\phi\rangle(r)$ and standard deviation $\sigma_\phi(r)$ of the potential of all (halo, particle) pairs in logarithmic shells of comoving distance r from 0.02 to $50\,h^{-1}\,$Mpc (36 bins), and the jackknife error of the mean over 3^3 sub-cubes of halo centres. Each code uses its own halos and its own particles. Only stacks are accumulated: per chunk of particles, a periodic KD-tree of the chunk is paired with trees of halo centres, and every pair adds $(1, \phi, \phi^2, r)$ to a (sub-cube, mass bin, shell) array, so memory does not depend on the number of halos or particles. To keep the cost per halo constant out to large radii the outer shells are diluted: shell i uses the nested subsample $u < 8^{-\ell_i}$ of the particles, with the smallest ℓ_i for which the expected number of particles inside the shell's outer radius is ≤ 16384. Here u is a hash of the particle ID (GADGET-4 ID -1 = PKDGRAV3 iOrder), uniform in $[0, 1)$ and independent of ϕ, so both codes use the same particles at every level and the subsampling noise cancels in their difference. (A first version drew u at random per chunk; its outer-shell noise, $\sim 2 \times 10^4\,(\mathrm{km\,s^{-1}})^2$, showed up as a spurious 3% code difference at $45\,h^{-1}\,$Mpc and as a difference between PKDGRAV3's HDF5 and `.pkc` stacks.) Each shell uses one level, so its mean and scatter are unbiased (checked against a brute-force calculation: identical without dilution, within $1.4\sigma_{\rm jk}$ with it). The pair counts also give the halo–matter cross-correlation $1 + \xi_{hm}(r)$. At 2048^3 the outermost level uses 8^{-5} of the particles, and the cost is $\sim n_{\rm halo}\times$ levels $\times 16384$ pairs plus one pass through the snapshot. The per-sub-cube sums are stored, so code differences can be given a *paired* jackknife error: leave the same sub-cube out of both codes and take the scatter of the difference. That removes the large-scale variance the two runs share, which dominates the error of each stack on its own. The figures are made by the notebook `stack_potential.ipynb`.

Volume-weighted stacks and linear bins. Counting every (halo, particle) pair makes the stack a *mass-weighted* average over the shell: dense parts (filaments, neighbouring halos) count more than voids. The *volume-weighted* mean and variance average over space instead (`--weight volume`). Each code's potential is put on an N_g^3 grid of cell centres (512^3, spacing $0.39\,h^{-1}\,$Mpc, at 256^3) by a weighted local linear fit $\phi \approx a + \vec{g}\cdot\Delta\vec{x}$ to the 16 nearest particles. The grid points are then stacked exactly like particles: same trees, jackknife, and dilution by grid index, so both codes use the same points. A plain inverse-distance average is biased toward the denser side of the neighbourhood, i.e. too deep near structures: on a synthetic clustered test it biased the stacked mean by up to 8%, against $\leq 1.8\%$ (median 0.4%) for the linear fit. The interpolation reads all particles into one KD-tree; beyond 512^3 particles it has to be done slab by slab (not yet implemented). Every stack is made with two binnings: 36 logarithmic bins from 0.02 to $50\,h^{-1}\,$Mpc, and 250 linear bins of $0.2\,h^{-1}\,$Mpc from 0 to $50\,h^{-1}\,$Mpc. Shells with fewer than 50 pairs or grid points are masked, and so are volume-weighted shells inside one grid cell, where the stack only sees the interpolation. Script: `run_stacks_local.sh` (run on the Mac from copies of the four snapshots).

Halo catalogues. `compare_halos.py` compares host-halo mass ($M_{\rm vir}$, M_{200b}) and $V_{\rm max}$ functions, and matches host halos one to one (mutual nearest centres within $0.5\,R_{\rm vir}$ of the GADGET-4 halo and $|\log_{10} M_{\rm pkd}/M_{\rm Gadget}| < 0.5$). For matched halos it reports, per mass bin, the median and 16–84% range of the ratios of $M_{\rm vir}$, M_{200b}, $V_{\rm max}$, concentration $R_{\rm vir}/r_s$, spin, shape and virial ratio, the centre offset and bulk-velocity difference, and the differences of the VAHC halo bias b_1 and tidal eigenvalues.

Per-halo potential from the pipeline (`.vpot`). After the merge of the `rockstar-pkc-reader` branch, VAHC runs as `vahc_fast.py` (`run_addvalue.py--fast--pot`). It reads every particle once and also writes `out_<snap>.vpot`: for each halo the mean particle potential within spheres of radius r_s, R_{200b} and 0.5, 1, $1.5\,R_{\rm vir}$ around the halo centre (with the particle counts), and the Gaussian-weighted mean at the seven VAHC scales ($2, 4, 6, 8\,R_{200b}$; 2, 3, $5\,h^{-1}\,$Mpc). The latter is $[W * \sum_i \phi_i\,\mathrm{CIC}_i]/[W * \sum_i \mathrm{CIC}_i]$ on the 512^3 VAHC grid at the halo's cell, so it is mass-weighted. The FITS stage merges these columns (`phi_rs` … `phi_R5Mpch`) into `out_<snap>_vahc.fits.gz` (header `HAS_VPOT`). PKDGRAV3: ϕ from the `.pkc` files with the self term removed

(header `self_term_kms2`). GADGET-4: `vpot_hdf5.py` reads `PartType1/Potential` from the HDF5 snapshot, $\phi =$ `Potential`$/a$ with no self term, and uses the same C routines (`vpot_sums.c`), so the two codes' `.vpot` are computed identically. The density and therefore `.vahc` do not change: rerun `.vahc` files are bit-identical to the previous ones for both codes (checked at $z = 0$), and the GADGET-4 sphere sums equal a KD-tree brute-force sum (no count differences, relative difference 10^{-7}). `compare_vpot.py` matches host halos as `compare_halos.py` does and reports, per output and mass bin, the median and 16–84% range of $\phi_{\rm pkd}/\phi_{\rm Gadget}$ and of $\phi_{\rm pkd} - \phi_{\rm Gadget}$ at every radius. Each code uses its own halo centre and radii, except for the fixed 2, 3, 5 h^{-1} Mpc. No zero point is removed.

Halo–halo potential cross-correlation. `scripts/post-process/halo_potential_xcorr.py` correlates the per-halo potentials of two mass bins A and B. For all pairs ($i \in A$, $j \in B$) at comoving separation r it measures the mean difference $\langle \phi_i - \phi_j \rangle(r)$, the structure function $\langle (\phi_i - \phi_j)^2 \rangle(r)$, the covariance and the correlation coefficient. Its default columns are ϕ within $R_{\rm vir}$, r_s, $1.5R_{\rm vir}$, $0.5R_{\rm vir}$ and R_{200b}. Bins are log or linear (range and number set on input), and every bin pair $A \le B$ is computed in the stacked-profile mass bins unless chosen otherwise. Errors are jackknife over 3^3 sub-cubes, by the position of halo i. All statistics follow from six weighted pair sums per column (N, $\sum \phi_i$, $\sum \phi_j$, $\sum \phi_i \phi_j$, $\sum \phi_i^2$, $\sum \phi_j^2$), which one pass over the pairs collects for all columns at once. The pass is written in C (`halo_pair_sums.c`), with OpenMP over the halos of A and the halos of B in a cell grid matched to the distance range. Nothing is stored per pair. For 2048^3 catalogues the large-r shells are diluted: a shell uses the A halos with $u(\mathrm{id}) < 2^{-l}$ (u a hash of the halo ID, independent of ϕ), with the smallest l that keeps it at about 2×10^7 pairs. Each shell then uses a single subsample, so it stays unbiased. *Checks*: the sums equal a numpy brute force to 10^{-13} (three bin pairs, all columns and jackknife regions, including halos without ϕ_{r_s}). Diluted and undiluted statistics agree within 0.4σ. The 256^3 catalogues take 0.3 s (30 log bins) and 0.6 s (250 linear bins). 2.5 million synthetic (unclustered) halos in a 1 h^{-1} Gpc box take 9 and 21 s on 8 threads with 1.2 GB. Clustering adds small-r pairs, which are not diluted.

Correlation multipoles in real and redshift space. `scripts/post-process/halo_xi_multipoles.py` measures the halo cross-correlation multipoles $\xi_\ell(s)$, $\ell = 0 \ldots 4$, between mass bins A and B. The odd multipoles, in particular the dipole, are included. The geometry is a periodic box with a fixed line of sight, so the plane-parallel approximation is exact. Four spaces are measured: *real* positions; *rsd*, where each halo moves along the line of sight by $v_\parallel/(aH)$; *grav*, which moves it by $v_g/(aH)$ with $v_g = -\phi/c$, the gravitational redshift of a source at potential ϕ seen from $\phi = 0$; and *rsd+grav*, which applies both. Displacements wrap periodically, and $H = 100E(a)\,\mathrm{km\,s^{-1}}/(h^{-1}\,\mathrm{Mpc})$. ϕ is `phi_rs`, the mean potential within r_s, which is closest to a central galaxy. Where it is undefined (an empty r_s sphere) `phi_R0p5vir` is used instead; both choices are options. A constant zero point of ϕ moves every halo alike and does not matter. Pairs are counted with Corrfunc `DDsmu` from the `adematti/Corrfunc` fork (branch `desi`), which counts signed $\mu = s_\parallel/s \in (-1, 1)$ in 200 bins, with $s = x_B - x_A$. A B population shifted by $+\delta$ along the line of sight therefore gives $\xi_1 = -\delta\, d\xi_0/ds$. The estimator is $\xi(s, \mu) = DD/RR - 1$, with RR analytic for the periodic box, and ξ_ℓ uses bin-integrated Legendre polynomials. Errors are jackknife over 3^3 regions of the A halo; all spaces use the same halos and regions. The potential's effect is therefore measured as a difference between spaces (rsd+grav − rsd, grav − real) with its own jackknife error, and most of the cosmic variance cancels. We use 2 h^{-1} Mpc bins from 0 to 100 h^{-1} Mpc and average the x, y and z lines of sight; each is also stored separately. *Checks*: $DD(s, \mu)$ equals a numpy brute force in every bin, in real and redshift space, for cross and auto pairs. A μ sign test and a periodic-wrap test were run on single pairs. A constant 1 h^{-1} Mpc shift of one bin gives $\xi_1 \simeq -\delta\, d\xi_0/ds$. Dipoles of same-bin pairs vanish identically, and the real-space quadrupole is zero within errors. Corrfunc with OpenMP gives bit-identical counts. On the Mac it needs `libomp`: Apple clang has no OpenMP, and gcc rejects Corrfunc's `-march=apple-m1`. With that, 2.5 million synthetic halos in 1 h^{-1} Gpc to 100 h^{-1} Mpc (2.2×10^{10} pairs) take 56 s per space and line of sight on 8 threads, with 0.8 GB.

4 Results at 32^3

The 32^3 run is the first test of the chain (particle mass $2.1 \times 10^{13}\,h^{-1}\,\mathrm{M}_\odot$, at most 15 halos per snapshot). Table 2 and the grey lines of Fig. 3 summarise the particle comparison.

out	z	corr(ϕ)	$\frac{\mathrm{rms}\,\Delta\phi}{\mathrm{rms}\,\phi}$	FFT_x	offset $(\mathrm{km\,s^{-1}})^2$	med $\lvert\Delta x\rvert$ $h^{-1}\,\mathrm{Mpc}$	90% $\lvert\Delta x\rvert$ $h^{-1}\,\mathrm{Mpc}$	med $\lvert\Delta v\rvert$ $\mathrm{km\,s^{-1}}$	corr(v)
0	11.98	0.99965	0.026	0.082	+1392	0.016	0.029	6.3	0.9989
19	2.96	0.99972	0.024	0.103	+451	0.098	0.167	13.3	0.9980
46	1.00	0.99520	0.098	0.117	+308	0.205	0.364	17.9	0.9633
64	0.50	0.98510	0.173	0.147	−419	0.271	0.531	20.8	0.9123
100	0.00	0.96891	0.253	0.220	−2011	0.379	0.891	24.9	0.8151

Table 2: 32^3, particle by particle. FFT_x: rms difference of the FFT potentials computed from the two codes' particle positions, over rms ϕ. Offset: median $\phi_{\mathrm{pkd}} - \phi_{\mathrm{Gadget}}$ after removing PKDGRAV3's self-term (-1.9×10^7 at $z = 12$, $-1.4 \times 10^6\,(\mathrm{km\,s^{-1}})^2$ at $z = 0$).

- **Potential.** Until $z \approx 3$ the two potentials agree to 2–3% rms (correlation 0.9997). Afterwards the difference grows to 25% at $z = 0$; the FFT potentials of the two codes' own particle positions differ by 22%, so most of the late-time difference comes from the particles being in different places, not from the force solvers. The zero points agree to $\lesssim 2 \times 10^3\,(\mathrm{km\,s^{-1}})^2$ (0.1% of the self-term that PKDGRAV3 carries).

- **Positions and velocities.** The median particle separation grows from $16\,h^{-1}$kpc at $z = 12$ to $0.38\,h^{-1}\,\mathrm{Mpc}$ at $z = 0$ ($1.8\,\epsilon_G$, 6% of the mean interparticle spacing $6.25\,h^{-1}\,\mathrm{Mpc}$), the median velocity difference from 6 to $25\,\mathrm{km\,s^{-1}}$. The rms velocity difference ($314\,\mathrm{km\,s^{-1}}$ at $z = 0$) is dominated by particles in halos, where trajectories are chaotic: there the median is small but the tail long (Fig. 1). The rms velocities of the two runs differ by 0.4% at $z = 12$ and 1.7% at $z = 0$.

- **Halos.** Rockstar finds 13 host halos in GADGET-4 and 8 in PKDGRAV3 at $z = 0$; 4 are matched one to one (the others merge or split differently at the 30–40 particle limit). Matched masses agree to a few per cent.

- **Stacked potential** (Fig. 2). Around the 9 halos above $4 \times 10^{14}\,h^{-1}\,\mathrm{M}_\odot$ the mean potential of the two codes agrees within the jackknife errors from 1 to $50\,h^{-1}\,\mathrm{Mpc}$; inside $\sim 0.5\,h^{-1}\,\mathrm{Mpc}$ PKDGRAV3's is deeper, as expected from its smaller effective softening (PKDGRAV3's force becomes Newtonian at $1.0\,\epsilon_G$, GADGET-4's at $2.8\,\epsilon_G$). With 1–9 halos per bin these numbers are indicative only.

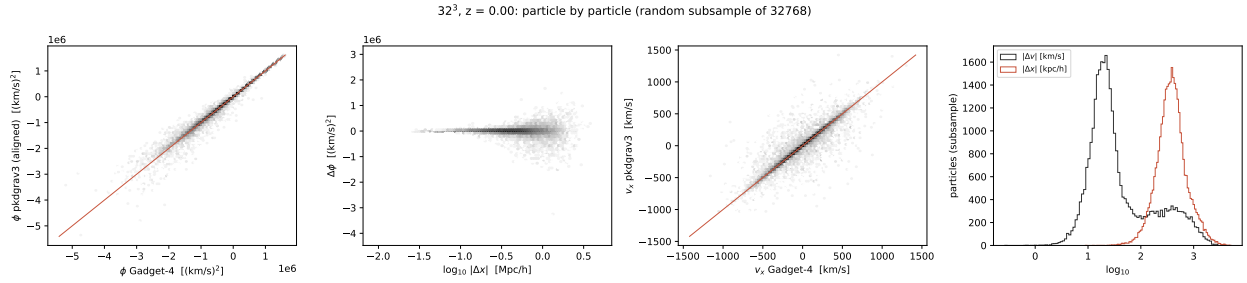


Figure 1: 32^3, $z = 0$, every particle: potentials (PKDGRAV3 aligned to GADGET-4's zero point), potential difference against particle separation, v_x, and the distributions of $|\Delta v|$ and $|\Delta x|$.

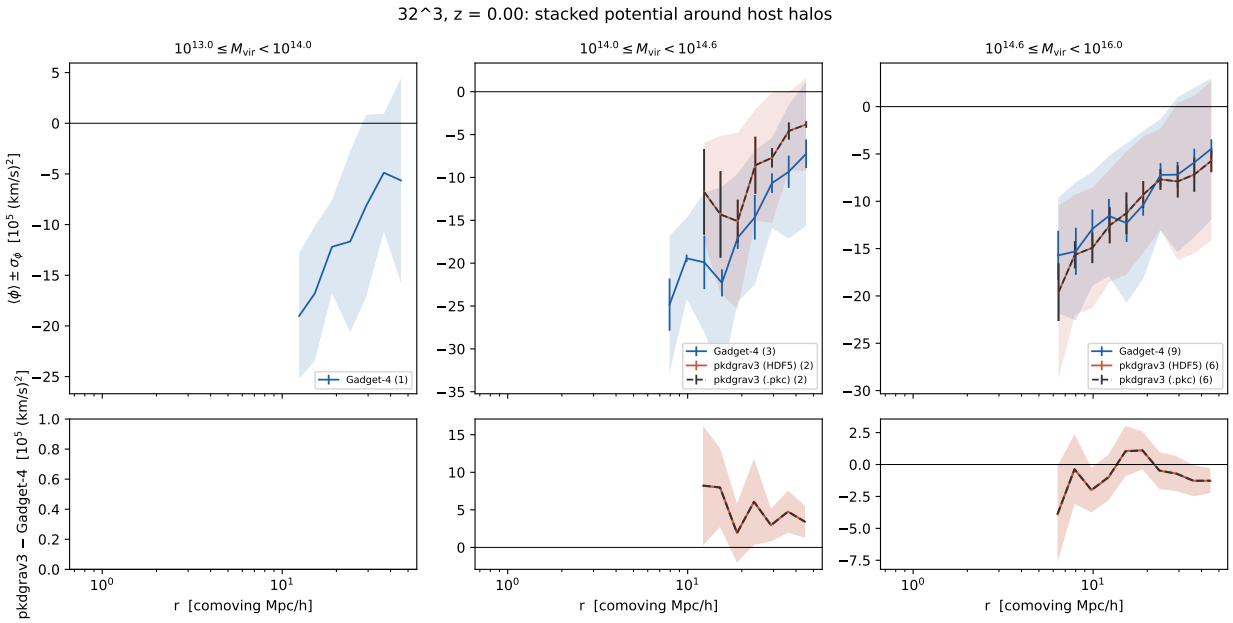


Figure 2: 32^3, $z = 0$: stacked potential around host halos in three $M_{\rm vir}$ bins (number of halos in the legend). Line with error bars: mean and jackknife error; band: $\pm\sigma_\phi$, the scatter of the potential over all pairs in the shell. Bottom: PKDGRAV3 − GADGET-4 (HDF5 and `.pkc` coincide).

The volume-weighted and linear-bin stacks were made at 32^3 too (64^3 grid, spacing $3.1\,h^{-1}$ Mpc). With 1–12 halos per bin and $0.2\,h^{-1}$ Mpc shells, almost every linear shell has fewer than 50 pairs and is masked. They are in `data/pkd32_vs_gadget32_r1/profiles` and the notebook, but carry no information beyond the 256^3 results.

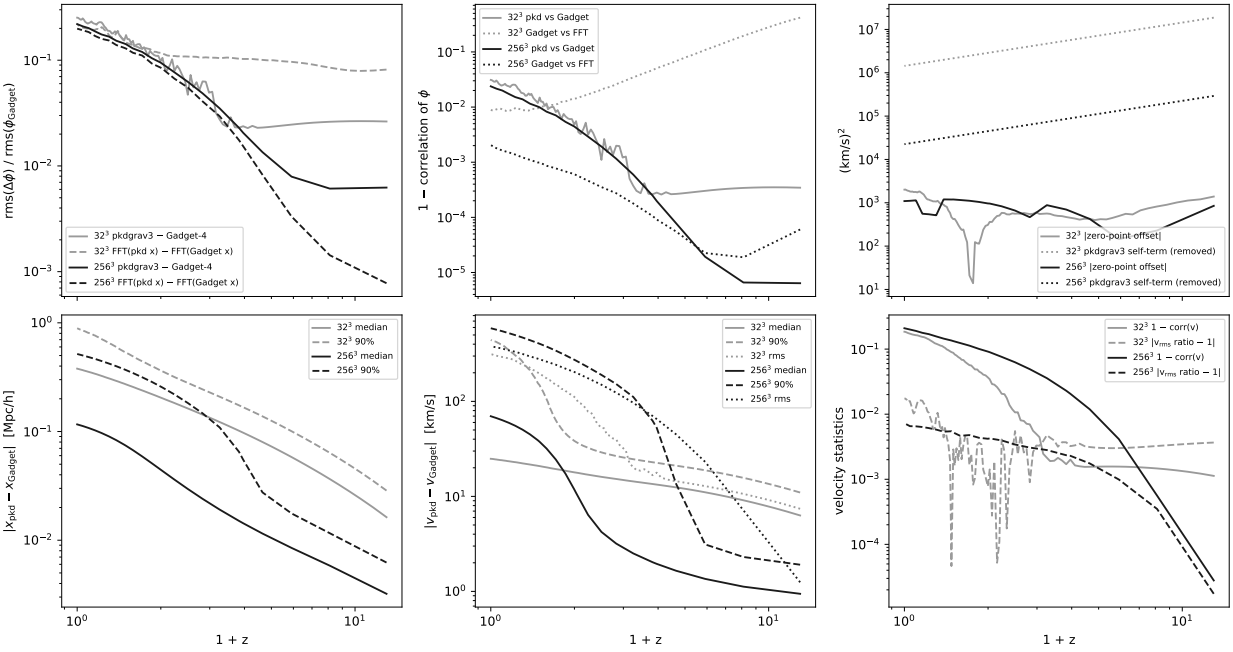


Figure 3: Particle-by-particle statistics against redshift (grey: 32^3, black: 256^3). Top: rms potential difference over rms potential (solid) and the same for the FFT potentials of the two codes' particle positions (dashed); 1−correlation of the potentials; zero-point offset and PKDGRAV3's removed self-term. Bottom: particle separation, velocity difference, velocity correlation and rms-velocity ratio.

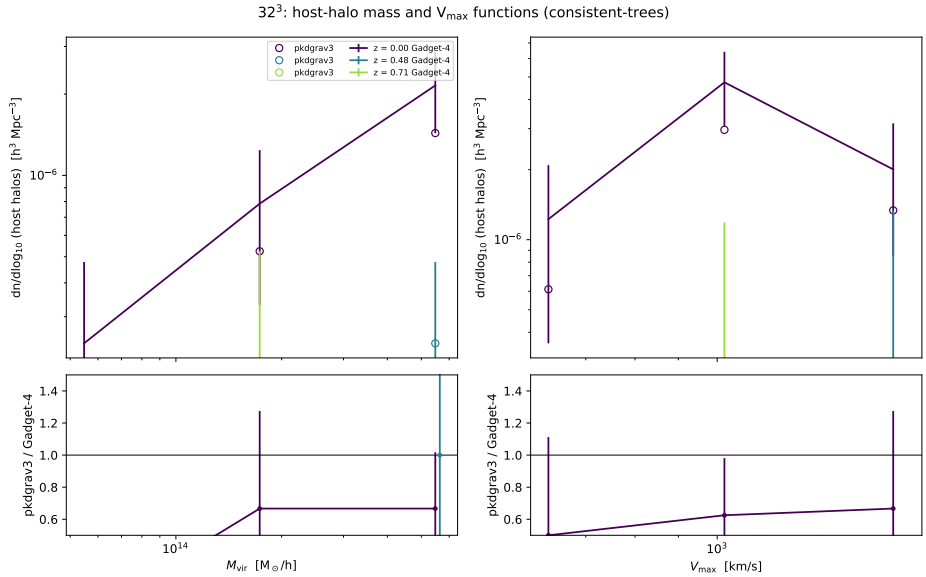


Figure 4: 32^3: host-halo $M_{\rm vir}$ and $V_{\rm max}$ functions (Rockstar) of both codes; with ≤ 15 halos per snapshot this is only a test of the tool.

5 Results at 256^3

The 256^3 run has particle mass $4.15 \times 10^{10}\,h^{-1}\,\mathrm{M}_\odot$, $\epsilon_G = 26\,h^{-1}$kpc and $\epsilon_P = 13\,h^{-1}$kpc. Both runs and the full halo pipeline completed (Table 3).

stage	GADGET-4	PKDGRAV3
simulation	2.9 h, 64 MPI ranks, 101 snapshots, 63 GB	2.6 h, 64 threads, 101 outputs, peak memory 14 GB
Rockstar	29 min, 40 658 halos at $z = 0$	29 min, 39 883 halos at $z = 0$
consistent-trees	10 min, 94 hlists ($z \leq 7.8$)	11 min, 94 hlists
VAHC (grid 512^3)	2.3 h with 8 processes (peak 168 GB)	3.5 h with 8 processes
comparison	29 min (23 outputs, peak 9 GB); stacks 3–13 s per code and output	

Table 3: 256^3 jobs on pawna. consistent-trees skips the 7 earliest catalogues (< 20000 bytes, production threshold), so at $z > 7.8$ the halo comparison uses the Rockstar catalogues.

5.1 Particles: potential, positions, velocities

out	z	corr(ϕ)	$\frac{\mathrm{rms}\,\Delta\phi}{\mathrm{rms}\,\phi}$	FFT$_x$	offset $(\mathrm{km\,s^{-1}})^2$	med $\lvert\Delta x\rvert$ h^{-1} Mpc	90% $\lvert\Delta x\rvert$ h^{-1} Mpc	med $\lvert\Delta v\rvert$ $\mathrm{km\,s^{-1}}$	corr(v)
0	11.98	0.999994	0.006	0.001	+849	0.0032	0.006	0.9	1.0000
19	2.96	0.999802	0.021	0.015	−640	0.014	0.055	2.0	0.979
46	1.00	0.995549	0.095	0.085	−966	0.045	0.259	11.9	0.908
64	0.50	0.990111	0.142	0.129	−1192	0.074	0.367	37.5	0.867
100	0.00	0.976212	0.220	0.199	−1093	0.116	0.517	69.8	0.791

Table 4: 256^3, particle by particle (columns as in Table 2). PKDGRAV3's removed self-term is $-2.9 \times 10^5\,(\mathrm{km\,s^{-1}})^2$ at $z = 12$ and -2.25×10^4 at $z = 0$.

- **Before structure forms the two codes agree very closely.** At $z = 12$ the potentials correlate to $1 - 6 \times 10^{-6}$ with 0.6% rms difference. That is better than either code agrees with the FFT reference ($1 - 6 \times 10^{-5}$), so it is limited by the reference, not by the codes. Particles are $3\,h^{-1}$kpc apart ($0.12\,\epsilon_G$) and velocities agree to $0.9\,\mathrm{km\,s^{-1}}$.

- **The difference then grows with nonlinear structure**, to 2% rms at $z = 3$, 10% at $z = 1$ and 22% at $z = 0$ (25% at 32^3). From $z \approx 3$ on, 75–90% of it is accounted for by the particles being in different places: the FFT potentials of the two sets of positions differ by 0.73 (at $z = 3$) to 0.90 (at $z = 0$) of the rms $\Delta\phi$ (Fig. 3). The force solvers themselves agree far better than the rms suggests (see also the matched halos below).

- **The zero points agree** to $\leq 1.2 \times 10^3\,(\mathrm{km\,s^{-1}})^2$ after removing PKDGRAV3's self-term, about 0.1% of rms ϕ. The `.pkc` potential matches PKDGRAV3's HDF5 to $0.5\,(\mathrm{km\,s^{-1}})^2$ (half the quantisation step) at every output.

- **Positions and velocities.** At $z = 0$ the median particle separation is $0.116\,h^{-1}\,\mathrm{Mpc}$ ($4.5\,\epsilon_G$; 15% of the mean interparticle spacing) and the 90th percentile $0.52\,h^{-1}\,\mathrm{Mpc}$. Velocities split into two populations (Fig. 5):

  – particles outside halos, $|\Delta v| \approx 3\,\mathrm{km\,s^{-1}}$;

  – particles inside halos, whose orbits are chaotic, $|\Delta v| \sim 100$–$300\,\mathrm{km\,s^{-1}}$.

  The median is $70\,\mathrm{km\,s^{-1}}$ and the rms $380\,\mathrm{km\,s^{-1}}$. The rms velocities of the two runs differ by 0.7% at $z = 0$.

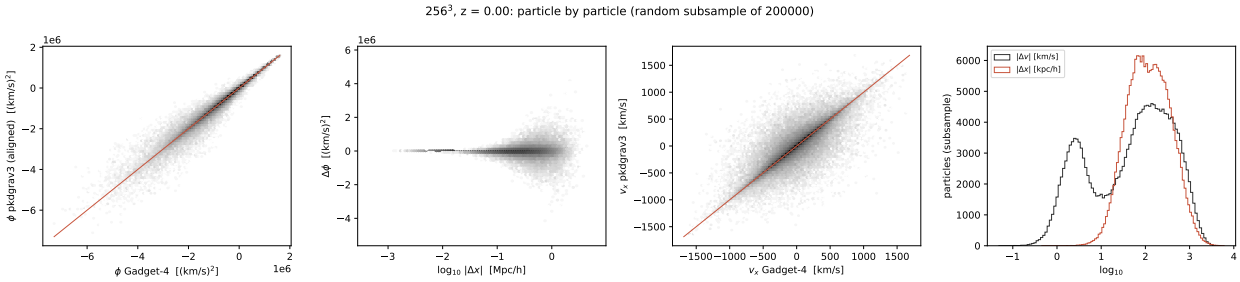


Figure 5: 256^3, $z = 0$, a random 200 000 particles: potentials, $\Delta\phi$ against separation, v_x, and the distributions of $|\Delta v|$ and $|\Delta x|$. The bimodal $|\Delta v|$ separates field particles from halo particles.

5.2 Halo catalogues and VAHC

At $z = 0$ there are 36 043 (GADGET-4) and 35 404 (PKDGRAV3) host halos; 28 484 are matched one to one.

- **Mass function** (Fig. 6). The M_{vir} functions agree within 2–3% (and within Poisson errors) from 10^{12} to $10^{14.5}\,h^{-1}\,\mathrm{M}_\odot$ at every redshift. Below $\sim 10^{12}\,h^{-1}\,\mathrm{M}_\odot$ (30 particles) PKDGRAV3 has 5% fewer halos at 5×10^{11} and 13% fewer at 2×10^{11}.

- **The V_{max} function differs more.** PKDGRAV3 has 14–17% fewer halos at 100–160 $\mathrm{km\,s^{-1}}$ and 13–19% more at 300–600 $\mathrm{km\,s^{-1}}$ ($z = 0$; about half as much at $z = 1$). This follows from the halo structure below.

- **Matched halos** (Fig. 7). The matched fraction is $\geq 96\%$ above $3 \times 10^{12}\,h^{-1}\,\mathrm{M}_\odot$, 79% at 10^{12}–3×10^{12} and 45% at 3×10^{11}–10^{12}.

  – *Masses* agree: median ratio 1.00 above $10^{12}\,h^{-1}\,\mathrm{M}_\odot$, with 16–84% range ±12% at 10^{12} narrowing to ±2% at 10^{14}.

  – PKDGRAV3 *halos are more concentrated*: R_{vir}/r_s is higher by 21–26% at 10^{12}–$10^{13}\,h^{-1}\,\mathrm{M}_\odot$, 9% at 3×10^{13} and 3% above 10^{14}. Their V_{max} is 2–5% higher.

  – *Shapes*: above $3 \times 10^{12}\,h^{-1}\,\mathrm{M}_\odot$ PKDGRAV3 halos are somewhat rounder, with c/a higher by 6–17%.

  – *Spin* agrees (median ratio 0.99–1.02).

– *Centres and bulk velocities*: the median centre offset is $0.24\,R_{\rm vir}$ at 10^{12}, 0.11 at 10^{13} and 0.03 at $3\times10^{14}\,h^{-1}\,{\rm M}_\odot$. Bulk velocities differ by 15–80 $\rm km\,s^{-1}$ (median).

The higher central density is consistent with PKDGRAV3's smaller effective softening ($2\epsilon_P=\epsilon_G$ against GADGET-4's spline support $2.8\epsilon_G$). It is largest at 10^{12}–$10^{13}\,h^{-1}\,{\rm M}_\odot$, where r_s is only a few ϵ_G.

- **VAHC.** The halo-by-halo bias b_1 is measured from the density field at $k<0.1\,h\,{\rm Mpc}^{-1}$, which the two runs share. It is identical (correlation 1.0000; rms difference 0.010, against a halo-to-halo scatter of 3.4). The tidal eigenvalue λ_1 correlates at 0.98 with $R=4R_{200b}$ smoothing (rms difference 1.0, scatter 5.4) and at 0.94 with $2R_{200b}$, where the halo's own surroundings, which differ in detail, matter more.

- **Potential of matched halos** (Fig. 8).

  – PKDGRAV3's $\phi_{\rm min}$ is 4–9% deeper (median per mass decade at $z=0$), the mean potential of the 10% most bound particles 4–6% deeper, and $\langle\phi\rangle$ within $R_{\rm vir}$ 1–2% deeper. This is the same central-density difference.

  – For the same particles, PKDGRAV3's mean potential of each GADGET-4 halo's members agrees with GADGET-4's to 0.1–0.5%. That is the cleanest test of the two potentials, free of halo-finder and centring differences.

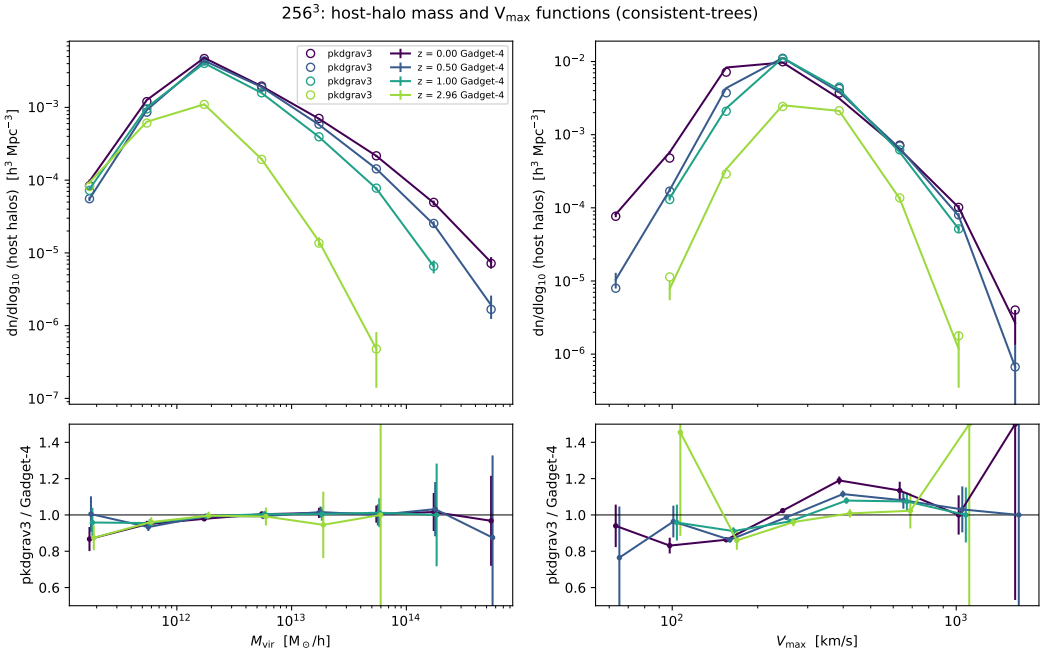


Figure 6: 256^3: host-halo $M_{\rm vir}$ and $V_{\rm max}$ functions from the consistent-trees catalogues of the two codes at $z\approx3$, 1, 0.5 and 0 (lines with errors: GADGET-4; circles: PKDGRAV3). Bottom: ratio with Poisson errors.

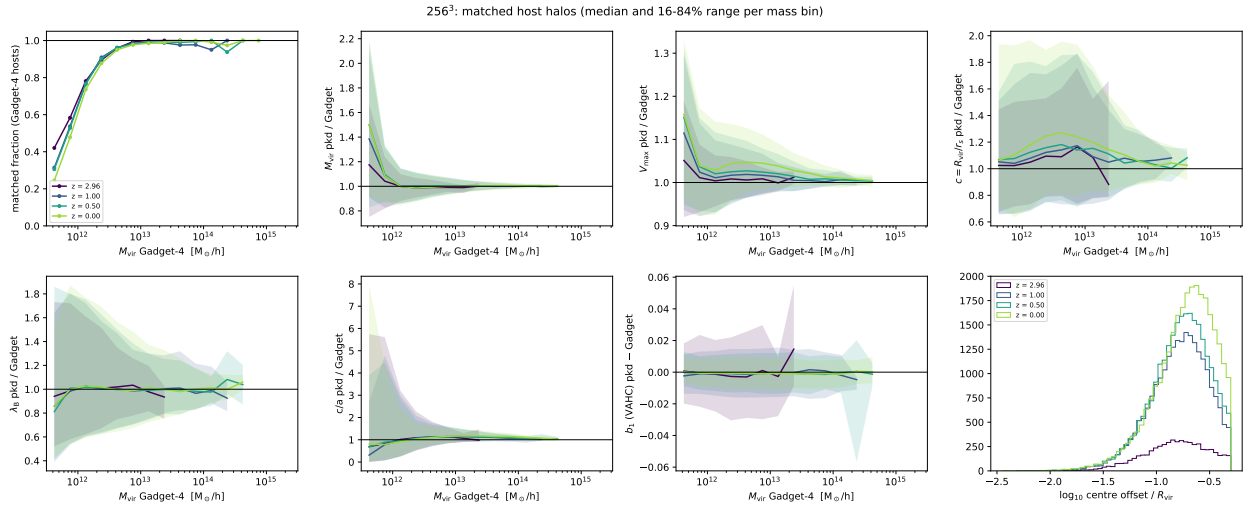


Figure 7: 256^3: matched host halos. Matched fraction, and per GADGET-4 mass bin the median (line) and 16–84% range (band) of PKDGRAV3/GADGET-4 for $M_{\rm vir}$, $V_{\rm max}$, concentration, spin and c/a; the difference of the VAHC bias b_1; and the distribution of centre offsets.

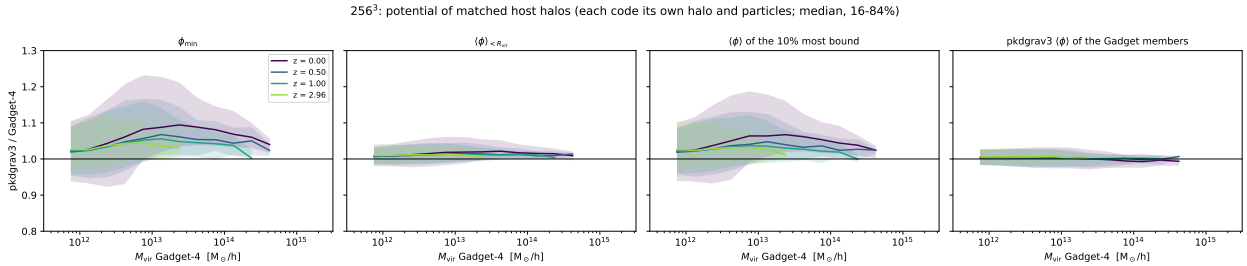


Figure 8: 256^3: potential of matched host halos, PKDGRAV3/GADGET-4. Each code uses its own halo and its own particles within $R_{\rm vir}$, except in the last panel, where PKDGRAV3's potential is evaluated for the GADGET-4 halo's member particles (same IDs).

5.3 Stacked potential around halos

Figures 9–12 come from the notebook `stack_potential.ipynb`. Each code uses its own host halos in four $M_{\rm vir}$ bins. At $z = 0$ these hold 13 375 / 3 181 / 525 / 78 halos in GADGET-4 (2×10^{12}–10^{13}, 10^{13}–5×10^{13}, 5×10^{13}–2×10^{14}, $> 2 \times 10^{14}\,h^{-1}\,{\rm M}_\odot$) and the same within 1% in PKDGRAV3. Differences carry the paired jackknife error: the two codes' differences over the same sub-cubes, which removes the shared large-scale (sample) variance. The dilution uses the same particle IDs in both codes, so the subsampling noise of the outer shells cancels as well.

- **Shape.** $\langle\phi\rangle(r)$ is deepest at the centre, from $-5 \times 10^5\,({\rm km\,s^{-1}})^2$ in the lowest bin to -5×10^6 in the highest. In the low-mass bins it has a second minimum at 1–3 h^{-1} Mpc: small halos often sit near larger ones. It is still $\approx -6 \times 10^5\,({\rm km\,s^{-1}})^2$ at $50\,h^{-1}$ Mpc, because halos live in large-scale potential wells. The scatter σ_ϕ in a shell, $(5\text{–}10) \times 10^5\,({\rm km\,s^{-1}})^2$, is set by the large-scale potential and is much larger than any code difference.
- **pkdgrav3 − Gadget-4.**
  - Inside $\sim 0.1\,h^{-1}$ Mpc PKDGRAV3 is deeper: by 3–5% at $0.05\,h^{-1}$ Mpc (3–6 paired errors), and by up to $3 \times 10^5\,({\rm km\,s^{-1}})^2$ (6%) at $0.02\,h^{-1}$ Mpc in the most massive bin. This is the softening and concentration difference.
  - From 0.2 to $50\,h^{-1}$ Mpc the mean potentials agree to 1–2% (mostly within 2 paired errors) at all four redshifts. In units of the shell scatter that is $|\Delta\langle\phi\rangle| \leq 0.02\,\sigma_\phi$.

- **`.pkc` = HDF5.** PKDGRAV3's compressed snapshots give the same stacks as its HDF5 snapshots at all radii.
- **Matter around the halos** (Fig. 12). $1+\xi_{hm}$ agrees within 2% beyond $0.1\,h^{-1}$ Mpc. Inside, PKDGRAV3 has more mass: +50% at $0.02\,h^{-1}$ Mpc in the lowest bin, falling to $\lesssim 10\%$ in the highest.

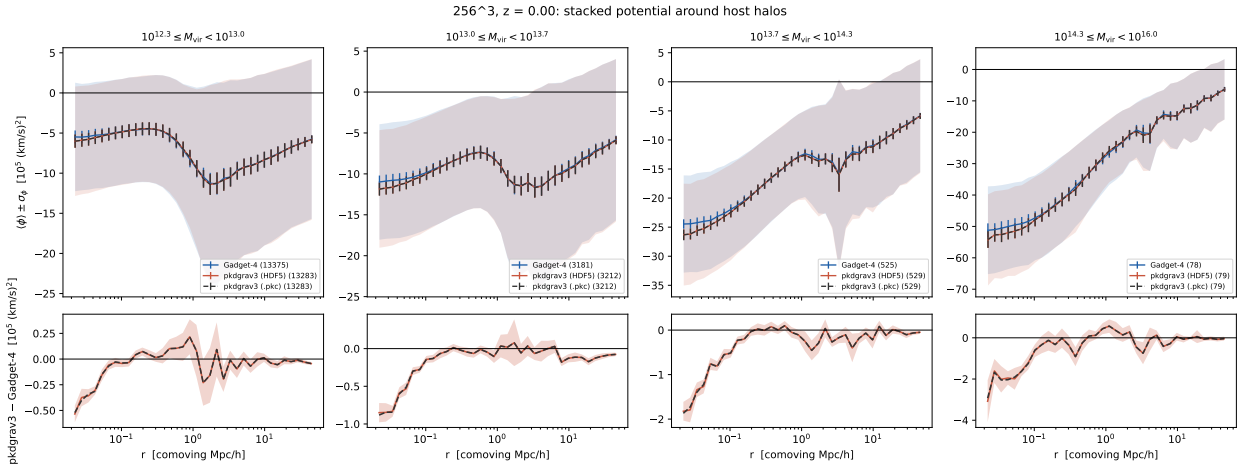


Figure 9: 256^3, $z = 0$: stacked potential around host halos in four $M_{\rm vir}$ bins out to $50\,h^{-1}$ Mpc (number of halos in the legend). Lines with error bars: mean and jackknife error; bands: $\pm\sigma_\phi$. Bottom: PKDGRAV3 − GADGET-4 (solid: HDF5, dashed: `.pkc`) with the paired jackknife error.

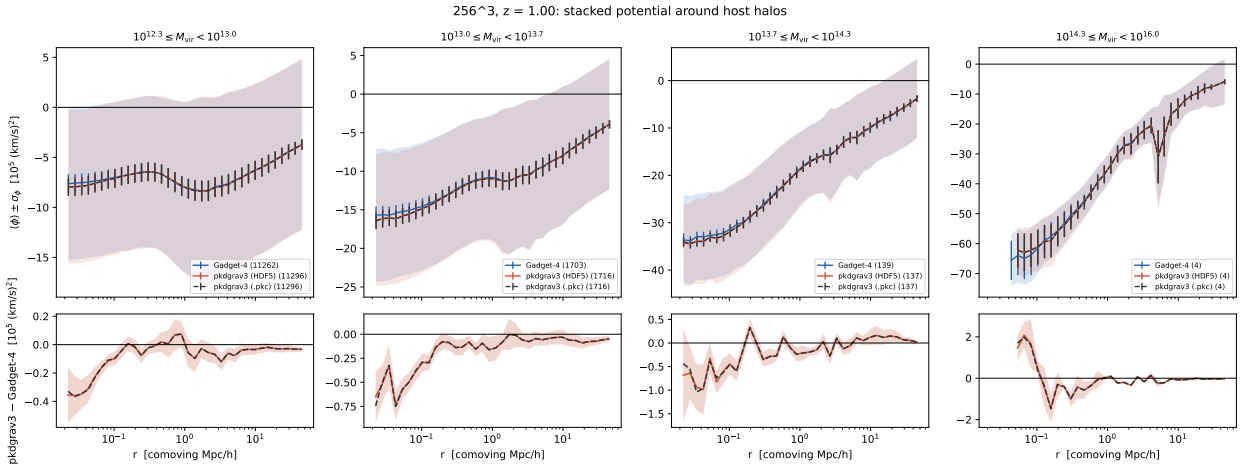


Figure 10: As Fig. 9 at $z = 1$.

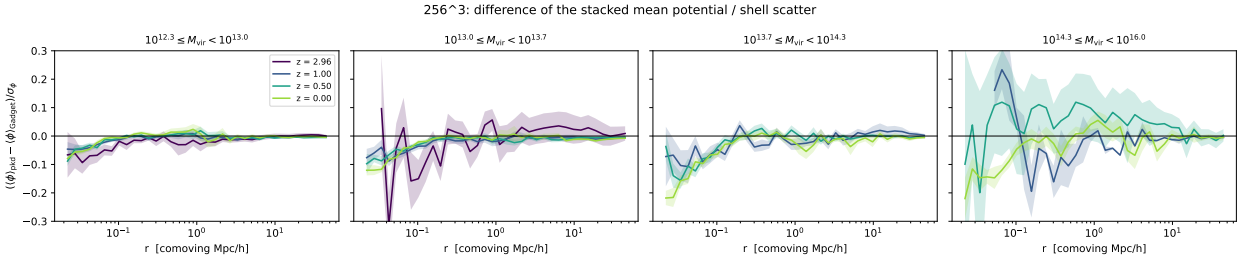


Figure 11: 256^3: $\Delta\langle\phi\rangle/\sigma_\phi$ at $z \approx 3$, 1, 0.5 and 0, with the paired jackknife error.

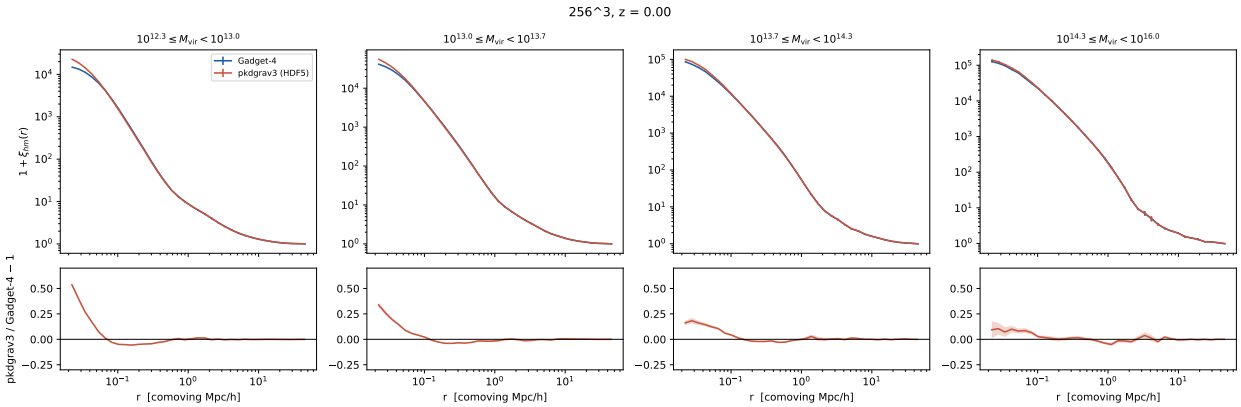


Figure 12: 256^3, $z = 0$: halo–matter cross-correlation from the same stacks; bottom: PKDGRAV3/GADGET-4 -1 with the paired jackknife error.

5.4 Volume-weighted stacks and linear bins

Figures 13–15 show the particle-weighted stacks (solid) and the volume-weighted stacks (dashed, 512^3 grid points) for both codes in the same mass bins. Table 5 lists them at $z = 0$ at selected radii of the $0.2\,h^{-1}$ Mpc linear bins. Volume-weighted shells inside one grid cell ($r < 0.39\,h^{-1}$ Mpc) are masked.

- **Volume versus particle weighting.** The volume-weighted mean is much shallower at every radius, and the gap grows outward. At $z = 0$ in the lowest bin it is $-3.1 \times 10^5\,(\mathrm{km\,s^{-1}})^2$ at $2\,h^{-1}$ Mpc (particle-weighted: -1.1×10^6) and -3×10^4 at $45\,h^{-1}$ Mpc (-5.7×10^5). Space far from a halo is mostly voids, where $\phi > 0$. The volume mean therefore tends to the box mean (ϕ averages to zero over the box), while particles keep sampling the wells of neighbouring structure. The volume-weighted scatter is also smaller, $(5.5\text{–}7) \times 10^5$ against $(7\text{–}13) \times 10^5\,(\mathrm{km\,s^{-1}})^2$, and it is almost flat in r. The bump of the particle-weighted stack at $1\text{–}3\,h^{-1}$ Mpc (neighbouring larger halos) is absent from the volume-weighted one, which rises smoothly.

- **pkdgrav3 − Gadget-4, volume-weighted.** For every mass bin with at least 10 halos, at all four redshifts and at all radii from 0.4 to $50\,h^{-1}$ Mpc, the difference is within 2 paired errors, and $|\Delta\langle\phi\rangle_V| \le 0.016\,\sigma_\phi$. The largest coherent offset is in the lowest bin at $z = 0$: PKDGRAV3 is shallower by 1.2–1.8 errors ($0.007\,\sigma_\phi$, about $4 \times 10^3\,(\mathrm{km\,s^{-1}})^2$) at all radii, which looks like a residual zero-point offset rather than a radial feature.

- **pkdgrav3 − Gadget-4, particle-weighted, linear bins.** The first shell (0–$0.2\,h^{-1}$ Mpc) holds the inner halo, and PKDGRAV3 is deeper there by 3–8 paired errors ($0.02\text{–}0.07\,\sigma_\phi$), as in the log bins. Beyond it most radii are within 2 errors. A few shells at 10–$45\,h^{-1}$ Mpc reach 3–5 errors, but they stay $\le 0.022\,\sigma_\phi$. The volume-weighted stacks do not show these excursions. They come from particles in neighbouring halos and filaments, whose positions differ between the codes (median $\Delta x = 0.12\,h^{-1}$ Mpc).

r [h^{-1} Mpc]	0.1	0.5	1.1	2.1	5.1	10.1	20.1	45.1
2×10^{12}–$10^{13}\,h^{-1}\,\mathrm{M}_\odot$ (13 375 halos)								
$\langle\phi\rangle_P$ [$10^5\,(\mathrm{km\,s^{-1}})^2$]	−4.9	−5.5	−9.4	−11.4	−9.3	−8.3	−7.0	−5.7
$\langle\phi\rangle_V$ [$10^5\,(\mathrm{km\,s^{-1}})^2$]	–	−3.5	−3.3	−3.1	−2.6	−2.0	−1.3	−0.34
Δ_P/err	−4.0	1.0	0.5	0.2	−0.2	−0.5	0.9	−2.9
Δ_V/err	–	1.8	1.4	1.6	1.7	1.8	1.8	1.2
10^{13}–$5 \times 10^{13}\,h^{-1}\,\mathrm{M}_\odot$ (3 181 halos)								
$\langle\phi\rangle_P$	−9.8	−7.4	−9.0	−11.4	−10.9	−9.1	−7.4	−5.8
$\langle\phi\rangle_V$	–	−6.2	−5.2	−4.6	−3.6	−2.8	−1.7	−0.47
Δ_P/err	−7.8	−0.8	0.4	−0.3	0.3	−2.4	−4.9	−3.2
Δ_V/err	–	−0.4	−0.8	−0.9	−0.8	−1.1	−0.3	−0.6
5×10^{13}–$2 \times 10^{14}\,h^{-1}\,\mathrm{M}_\odot$ (525 halos)								
$\langle\phi\rangle_P$	−21.7	−15.4	−12.4	−13.3	−11.9	−10.3	−8.4	−6.0
$\langle\phi\rangle_V$	–	−13.5	−10.1	−8.0	−5.9	−4.4	−2.7	−0.67
Δ_P/err	−7.0	0.5	−1.0	0.2	−1.0	−1.0	0.9	0.0
Δ_V/err	–	−0.8	−0.1	−0.4	0.0	0.1	0.4	0.8
$> 2 \times 10^{14}\,h^{-1}\,\mathrm{M}_\odot$ (78 halos)								
$\langle\phi\rangle_P$	−46.5	−34.6	−26.6	−21.1	−16.1	−13.4	−10.1	−6.1
$\langle\phi\rangle_V$	–	−30.6	−22.2	−16.3	−10.6	−7.6	−4.5	−1.2
Δ_P/err	−2.8	−0.5	1.8	0.9	−0.4	0.3	−0.2	−1.7
Δ_V/err	–	−0.6	0.1	0.2	−0.1	0.0	0.9	0.4

Table 5: 256^3, $z = 0$, linear $0.2\,h^{-1}$ Mpc bins (shell centres): GADGET-4 particle-weighted (P) and volume-weighted (V) mean potential, and PKDGRAV3 − GADGET-4 over its paired jackknife error. The full table for all redshifts is printed in `stack_potential.ipynb`.

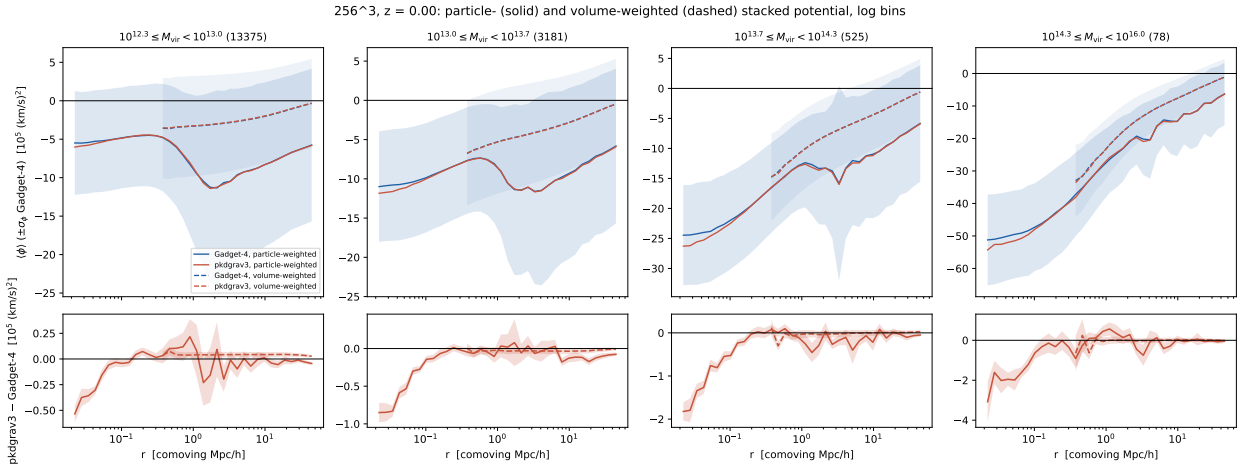


Figure 13: 256^3, $z = 0$, log bins: particle-weighted (solid) and volume-weighted (dashed) stacks for GADGET-4 and PKDGRAV3. Bands: $\pm\sigma_\phi$ of GADGET-4 for each weighting (the narrower band, starting at $0.4\,h^{-1}$ Mpc, is volume-weighted). Bottom: PKDGRAV3 − GADGET-4 with the paired jackknife error.

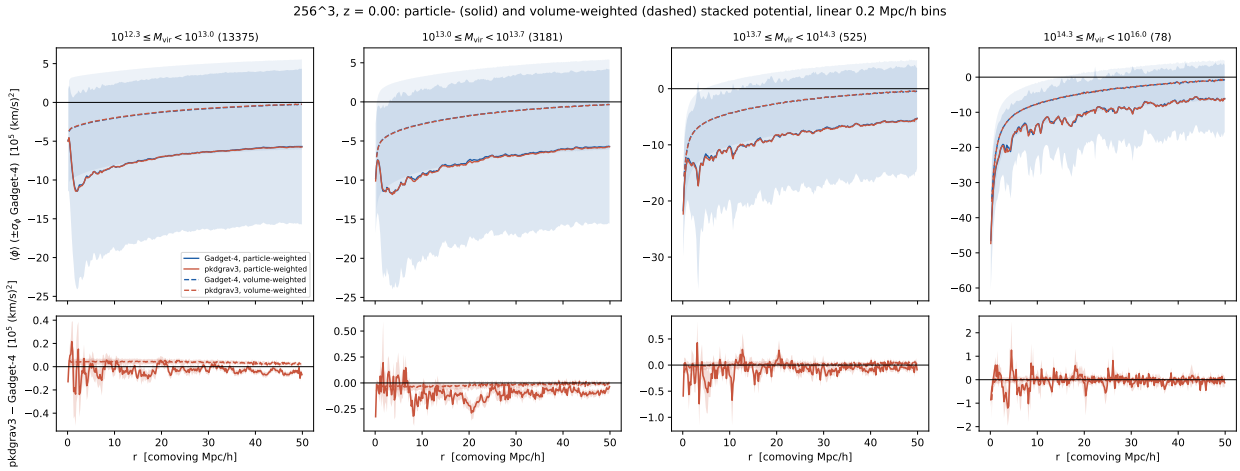


Figure 14: As Fig. 13 in 250 linear bins of $0.2\,h^{-1}$ Mpc from 0 to $50\,h^{-1}$ Mpc.

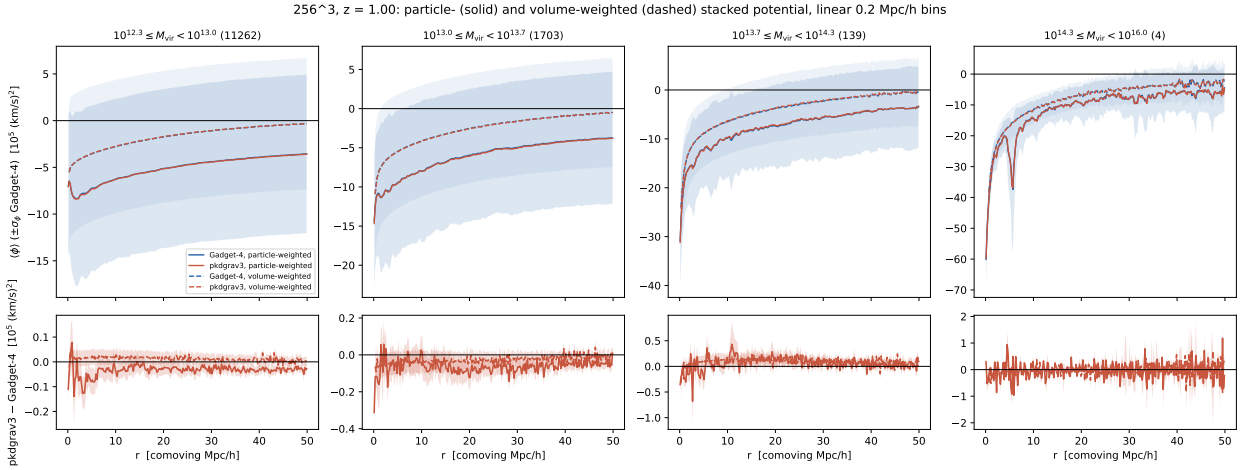


Figure 15: As Fig. 14 at $z = 1$.

5.5 Per-halo potential from the pipeline (`.vpot`)

VAHC and FITS were rerun for both 256^3 runs with `vahc_fast.py` and `.vpot` (Section 3; pawna jobs 2536704/2536706, FITS 2536792/2536793). All 94 outputs of each code have `.vahc` and `.vpot`, and `out_<snap>_vahc.fits.gz` carries the twelve `phi_*` columns. The previous files are in `<halos>/vahc_prev/`. The 32^3 runs cannot be redone because VAHC needs the consistent-trees catalogues. Table 6 and Figs. 16 and 17 compare matched host halos (28 484 at $z = 0$).

- **pkdgrav3 is deeper at every radius, by about 1% far from the centre.** The median excess grows inwards: at $z = 0$ it is 1.0–1.3% at $5\,h^{-1}$ Mpc, 0.9–1.6% at $4R_{200b}$, 1.3–2.1% within $R_{\rm vir}$ and 3–6% within r_s (above $10^{12}\,h^{-1}\,\mathrm{M}_\odot$). The $\langle\phi\rangle_{<R_{\rm vir}}$ excess agrees with the particle-level halo comparison above (Fig. 8: 1–2%), and the r_s excess is the higher concentration of PKDGRAV3 halos. Halos below $10^{12}\,h^{-1}\,\mathrm{M}_\odot$ (fewer than 100 particles) are 0.7–1.0% deeper at every radius.

- **The excess within r_s grows with time**, from ∼1.7% at $z = 3$ to 3–6% at $z = 0$ (Fig. 17), as the concentration difference does. At $5\,h^{-1}$ Mpc it grows from 0.5% to 1%.

- **The outer offset is a few 10^3 $(\mathrm{km\,s^{-1}})^2$ and is mass-weighted.** At $5\,h^{-1}$ Mpc the median difference is -6.5×10^3 to $-9\times10^3\,(\mathrm{km\,s^{-1}})^2$ at $z = 0$ (-1.9×10^4 above $10^{14}\,h^{-1}\,\mathrm{M}_\odot$). The particle zero points differ by only -1.1×10^3 (median over all particles), but the *mean* over all particles of $\phi_{\rm pkd} - \phi_{\rm Gadget}$ is $-8.6\times10^3\,(\mathrm{km\,s^{-1}})^2$. The `.vpot` means are particle (mass)

averages, and the particles in the halo cores inside any sphere sit deeper in PKDGRAV3's more concentrated halos. So the outer offset is the same core difference, not a zero-point or unit problem. The volume-weighted stacks, which do not weight the halo cores, agree within errors (Section 3).

- **Halo to halo** the ratio scatters by ±3–4% within $R_{\rm vir}$ and ±1.5% at $5\,h^{-1}\,$Mpc (16–84%, 10^{12}–$10^{13}\,h^{-1}\,\mathrm{M}_\odot$, $z = 0$), more than the median offsets. This scatter comes from the halo finder and the particles ending up in different places.

$M_{\rm vir}$ (GADGET-4) $[h^{-1}\,\mathrm{M}_\odot]$ matched hosts with `.vpot`	10^{11}–10^{12} 2402	10^{12}–10^{13} 22 335	10^{13}–10^{14} 3531	$> 10^{14}$ 214
within r_s	$+0.8\,[-3.0, +6.2]$	$+3.1\,[-4.9, +13.4]$	$+6.1\,[+0.1, +15.9]$	$+3.1\,[-0.1, +7.3]$
within $R_{\rm vir}$	$+0.8\,[-1.9, +4.1]$	$+1.3\,[-2.2, +5.3]$	$+2.1\,[-0.4, +5.8]$	$+1.5\,[+0.1, +3.0]$
$4R_{200b}$ (Gaussian)	$+0.7\,[-1.4, +3.5]$	$+0.9\,[-1.7, +3.9]$	$+1.6\,[-0.4, +4.5]$	$+1.4\,[+0.3, +2.6]$
$5\,h^{-1}\,$Mpc (Gaussian)	$+1.0\,[-0.7, +2.7]$	$+1.0\,[-0.7, +2.6]$	$+1.2\,[-0.2, +2.6]$	$+1.3\,[+0.6, +2.2]$
$\Delta\phi$ at $5\,h^{-1}\,$Mpc $[10^3\,(\mathrm{km\,s^{-1}})^2]$	-6.5	-6.7	-9.2	-19.2

Table 6: 256^3, $z = 0$: $\phi_{\rm pkd}/\phi_{\rm Gadget} - 1$ in % of the pipeline `.vpot` potential of matched host halos, median [16%, 84%]. The number of hosts with r_s values is smaller (479, 14 501, 3528, 214), because the sphere within r_s is empty for some halos. All outputs: `data/pkd256_vs_gadget256_r1/vpot/vpot_match.csv`.

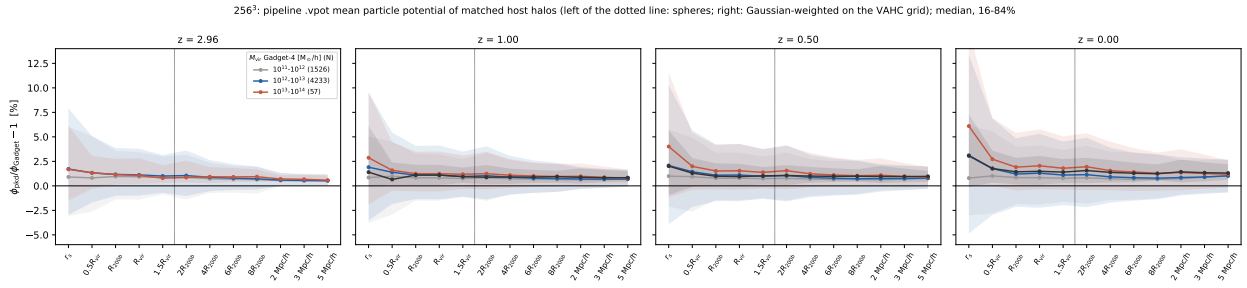


Figure 16: 256^3: pipeline `.vpot` potential of matched host halos, $\phi_{\rm pkd}/\phi_{\rm Gadget} - 1$ against radius at $z \approx 3$, 1, 0.5 and 0, per GADGET-4 mass bin (median and 16–84% band; N matched hosts). Left of the dotted line: spheres around each code's own halo centre with its own radii. Right: Gaussian-weighted means on the VAHC grid.

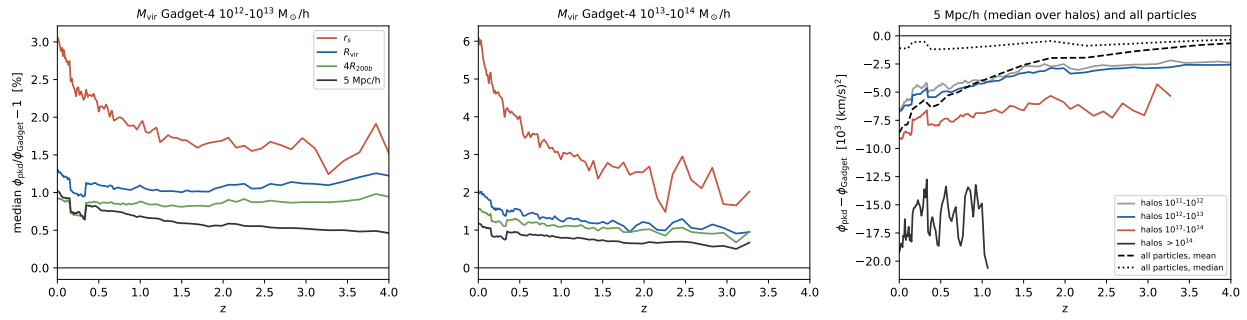


Figure 17: 256^3, all 94 outputs. Left and middle: median $\phi_{\rm pkd}/\phi_{\rm Gadget} - 1$ against z within r_s and $R_{\rm vir}$ and at $4R_{200b}$ and $5\,h^{-1}\,$Mpc. Right: median $\phi_{\rm pkd} - \phi_{\rm Gadget}$ at $5\,h^{-1}\,$Mpc per mass bin, with the mean (dashed) and median (dotted) of $\phi_{\rm pkd} - \phi_{\rm Gadget}$ over all particles (every 5th output). The halo values follow the particle mean, not the median zero point.

5.6 Halo–halo potential cross-correlation

Figures 18–20 show $z = 0$ for both codes, in the four stacked-profile mass bins (0: 2×10^{12}–10^{13}, 1: 10^{13}–5×10^{13}, 2: 5×10^{13}–2×10^{14}, 3: $> 2 \times 10^{14}\,h^{-1}\,\mathrm{M}_\odot$; 13 375, 3181, 525 and 78 GADGET-4

hosts) and all ten bin pairs. No shell needed dilution at this size. The outputs are on pawna in `data/compare/pkd256_vs_gadget256_r1/xcorr/` and reduced copies are in the report's `data/pkd256_vs_gadget256_r1/xcorr/`.

- **The two codes agree.** For $\phi_{R_{\rm vir}}$, PKDGRAV3 − GADGET-4 is within 2.5 jackknife errors (added in quadrature) in every shell with at least 20 pairs. It is within 1σ above $2\,h^{-1}$ Mpc, for every bin pair and every statistic. The largest deviations ($\sim 2\sigma$) are in the least massive auto pair below $2\,h^{-1}$ Mpc, which has few pairs.

- **Mean difference between bins** (Fig. 18, left). Lower-mass halos sit in shallower potentials. $\langle\phi_i - \phi_j\rangle$ for bins 0 and 3 grows from 1.1×10^6 $(\mathrm{km\,s^{-1}})^2$ at $2\,h^{-1}$ Mpc to 2.6×10^6 at $50\,h^{-1}$ Mpc; for bins 0 and 1 it grows from 0.1×10^6 to 0.35×10^6. It rises with r because a close pair shares the potential of its common environment: a low-mass halo next to a cluster is deeper than the average low-mass halo.

- **Structure function and correlation within a bin** (Fig. 18, middle and right). For bin 0, $\langle(\phi_i - \phi_j)^2\rangle^{1/2}$ grows from 2×10^4 $(\mathrm{km\,s^{-1}})^2$ at $0.4\,h^{-1}$ Mpc to 1.2×10^5 at 2, 4.8×10^5 at 18 and 7.8×10^5 at $50\,h^{-1}$ Mpc. The correlation coefficient falls from 0.999 to 0.98, 0.92, 0.72, 0.49 and 0.28 at 0.4, 2, 6, 18, 32 and $50\,h^{-1}$ Mpc. In linear bins (Fig. 19) it falls almost linearly with r out to $50\,h^{-1}$ Mpc: the halo potentials are coherent on the scales of the large-scale potential field.

- **The ϕ columns** (Fig. 20). Within one bin the five columns are almost equally correlated: the large-scale potential dominates, and the halo's own well adds a nearly constant offset. Between bins 0 and 3, ϕ_{r_s} is the least correlated column and has the largest mean difference, because it includes the inner profile of the massive halos. ϕ within $0.5R_{\rm vir}$ lies between r_s and $R_{\rm vir}$, and $1.5R_{\rm vir}$ and R_{200b} are the smoothest.

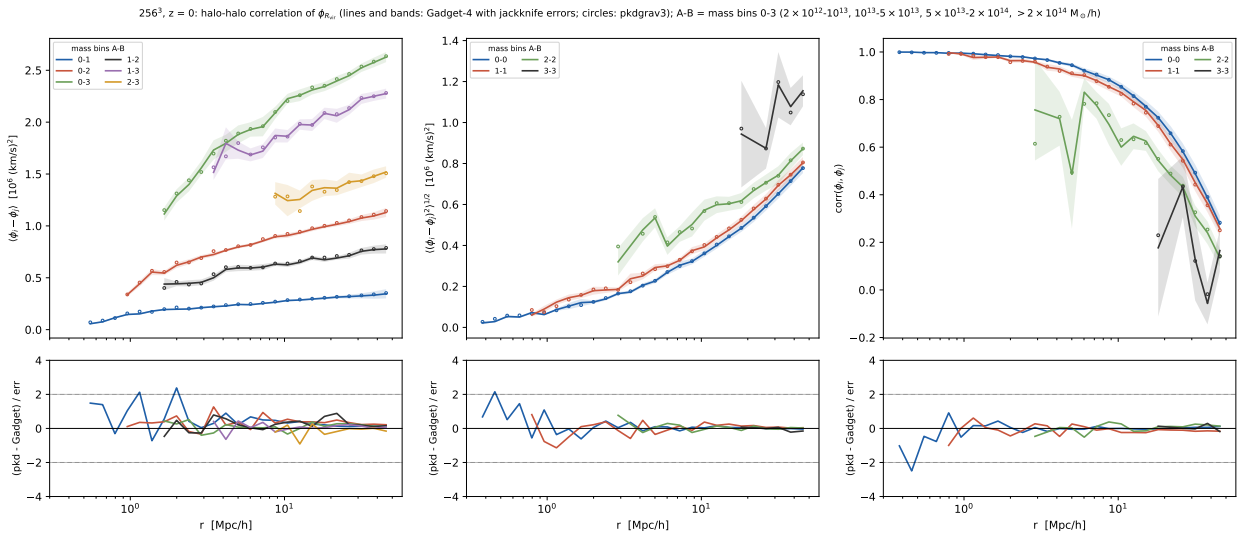


Figure 18: 256^3, $z = 0$: halo–halo correlation of $\phi_{R_{\rm vir}}$ in 30 log bins. Left: mean difference between different mass bins A–B (ϕ_i in A, ϕ_j in B). Middle and right: rms difference and correlation coefficient within a bin. Lines and bands: GADGET-4 with its jackknife error; circles: PKDGRAV3. Bottom: (PKDGRAV3 − GADGET-4) over the two errors in quadrature. Shells with fewer than 20 pairs are omitted.

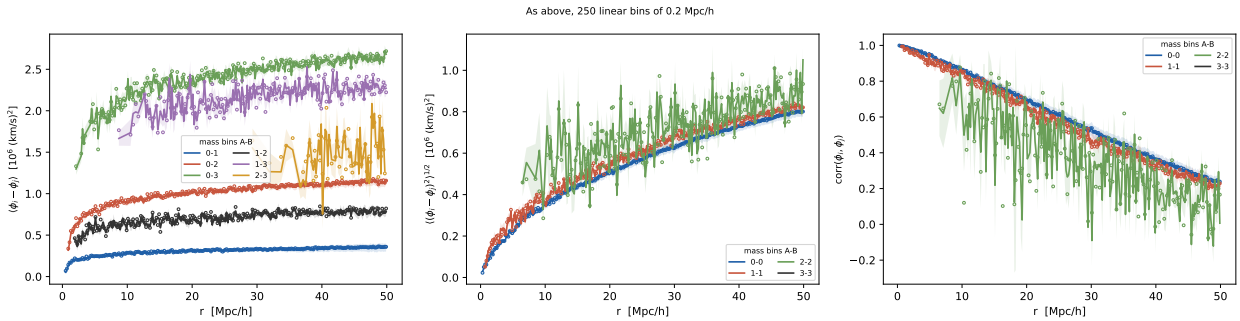


Figure 19: As Fig. 18 (top row) in 250 linear bins of $0.2\,h^{-1}$ Mpc.

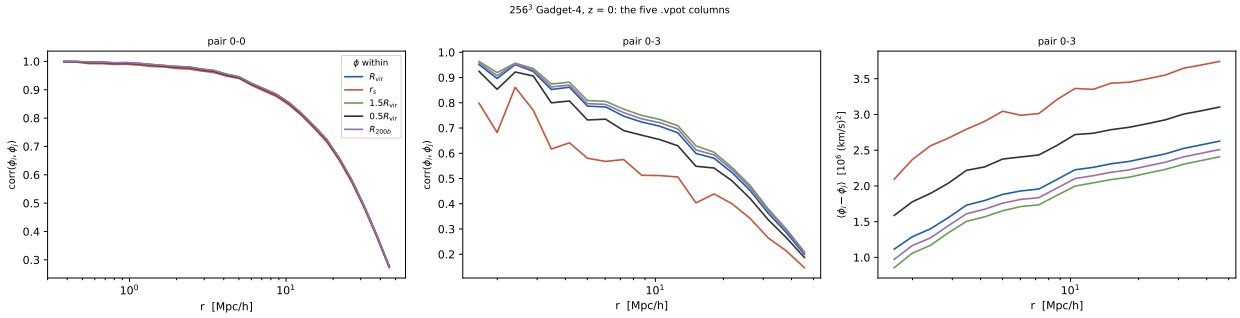


Figure 20: 256^3 GADGET-4, $z = 0$: the five ϕ columns. Correlation coefficient within bin 0 and between bins 0 and 3, and the mean difference between bins 0 and 3.

5.7 Correlation multipoles in real and redshift space

Figures 21 and 22 show $z = 0$ for both codes and all ten bin pairs. They use $2\,h^{-1}$ Mpc bins out to $100\,h^{-1}$ Mpc, the mean over the x, y and z lines of sight, and `phi_rs` for the potential shift. About 430 of the ∼36 000 hosts have neither ϕ_{r_s} nor $\phi_{0.5R_{\rm vir}}$ and are left out; all of them lie below the lowest mass bin. The outputs (on pawna in `data/compare/pkd256_vs_gadget256_r1/ximult/`) took 15 s per code on the Mac.

- **The codes agree.** PKDGRAV3 − GADGET-4 over the two jackknife errors in quadrature has an rms of 0.2 (ξ_0) and 0.5–0.9 (ξ_2) over all bin pairs and $s < 60\,h^{-1}$ Mpc, in real and redshift space. The largest single deviation is 3.3σ, in one of 300 real-space ξ_2 bins.

- **Real space**: the quadrupole and hexadecapole are zero within errors, as expected.

- **Redshift space** shows the Kaiser effect. For bin 0, $\xi_0^{\rm rsd}/\xi_0^{\rm real} \simeq 1.35$–$1.4$ at 10–$30\,h^{-1}$ Mpc, which corresponds to $\beta \simeq 0.5$ (for $f(z{=}0) \simeq 0.53$, a bias $\simeq 1.1$). The quadrupole is negative: $\xi_2/\xi_0^{\rm real}$ goes from -0.7 to -1.2 over the same range. Without the potential, the redshift-space dipole of cross pairs is consistent with zero, as it must be in the plane-parallel limit. For same-bin pairs it vanishes identically.

- **The dipole from the potential is small but measured.** The mean $v_g = -\phi_{r_s}/c$ per mass bin is 1.7, 3.4, 7.2 and 14.6 km s^{-1}, against RSD velocities of hundreds of km s^{-1}; it shifts halos by only 0.02–$0.15\,h^{-1}$ Mpc. The difference $\xi_1({\rm rsd}+{\rm grav}) - \xi_1({\rm rsd})$ (Fig. 22) is positive, as expected for a more massive B, for the pairs 0-1, 0-2, 0-3, 1-2 and 1-3 in both codes. Below $50\,h^{-1}$ Mpc (five 10-h^{-1} Mpc bins) χ^2 against zero is 62–170 for 0-1, 69–78 for 0-2 and 23–26 for 0-3. Covariances between bins are ignored here, so these values overstate the significance somewhat. In units of s^2, the mean amplitude is 0.06 (0-1), 0.3–0.4 (0-2) and 0.8–1.1 (0-3).

- **Amplified test.** With the potential shift $\times 50$, the measured dipole is 1.17, 0.94 and 0.54 times the simple prediction $-(\langle\delta_B\rangle - \langle\delta_A\rangle)\, d\xi_0/ds$ for 0-1, 0-2 and 0-3 at $s < 30\,h^{-1}\,$Mpc. It falls short of the prediction for close pairs of very different masses: neighbouring halos share their environment's potential, so they differ less than the bin means do (Section 5.6). Divided by 50 it matches the physical amplitudes above, so the response is linear and the physical measurement is the gravitational-redshift dipole itself.

- At 256^3 in $200\,h^{-1}\,$Mpc the dipole is at the $\sim 1\sigma$ level per $10\,h^{-1}\,$Mpc bin. At 2048^3 (more halos, and lower masses with smaller v_g but many more pairs) the same measurement is a stage of the pipeline (`ximult`).

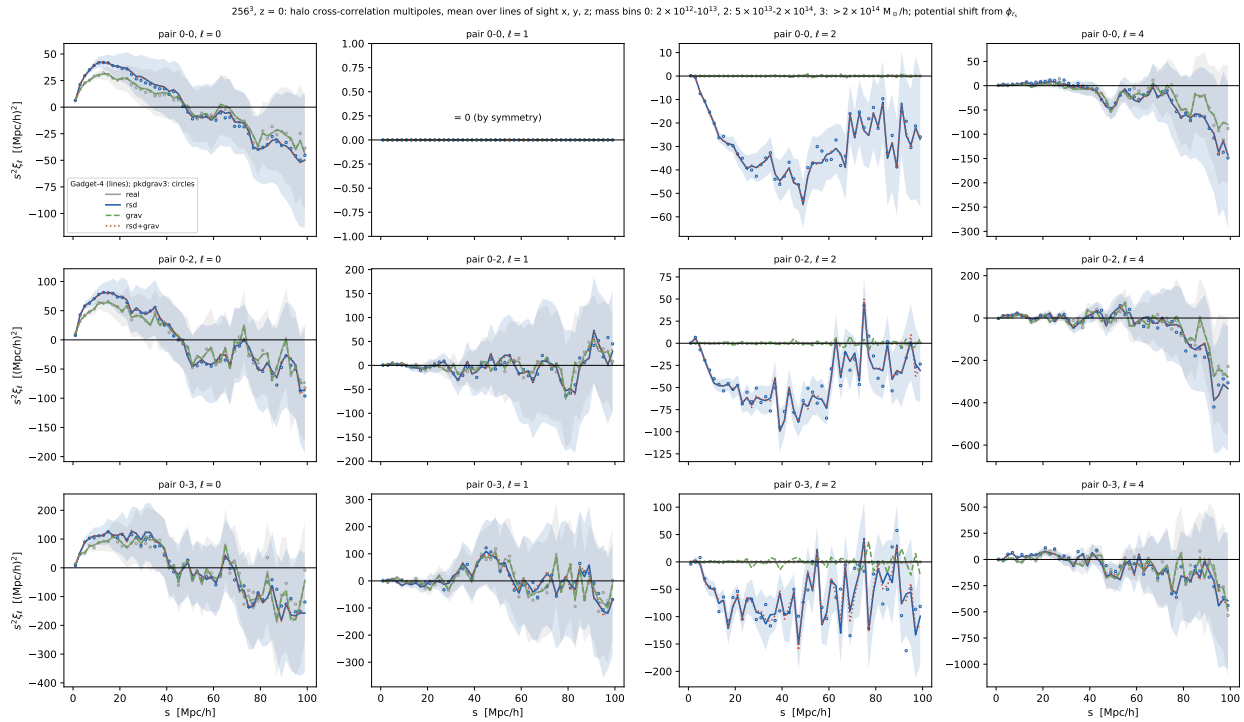


Figure 21: 256^3, $z = 0$: $s^2\xi_\ell$ for $\ell = 0, 1, 2, 4$ and bin pairs 0-0, 0-2 and 0-3 in the four spaces (mean over the x, y and z lines of sight). Lines: GADGET-4 (bands: jackknife errors of real and rsd). Circles: PKDGRAV3 (real and rsd). The same-bin dipole is zero by construction.

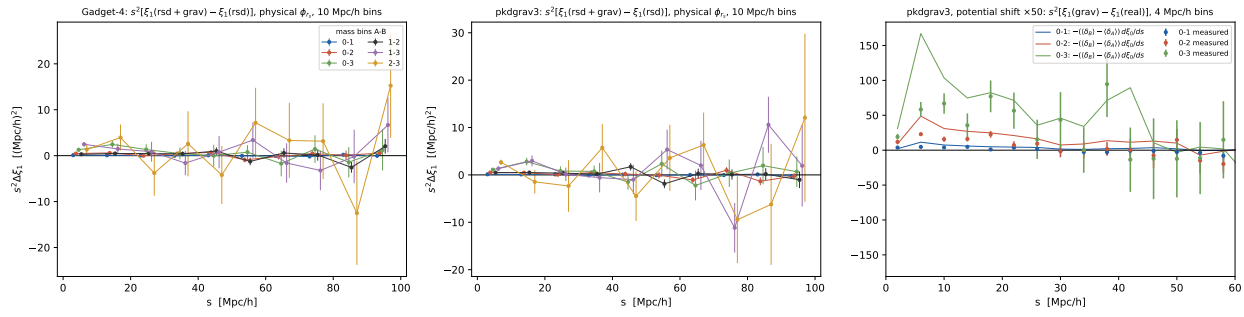


Figure 22: 256^3, $z = 0$: the dipole from the halo potential. Left and middle: $s^2[\xi_1(\mathrm{rsd} + \mathrm{grav}) - \xi_1(\mathrm{rsd})]$ for the cross pairs, in $10\,h^{-1}\,$Mpc bins (volume-weighted combinations of the $2\,h^{-1}\,$Mpc bins) with jackknife errors. Right: PKDGRAV3 with the potential shift $\times 50$ (real space), measured (points, $4\,h^{-1}\,$Mpc bins) against $-(\langle\delta_B\rangle - \langle\delta_A\rangle)\, d\xi_0/ds$ (lines).

6 Problems found on the way

1. **Gadget-4 writes Potential = 0 without `EVALPOTENTIAL`.** The first 32^3 run, with the exact production Config (`OUTPUT_POTENTIAL` only), has zero potential everywhere (`data_run1_zero pot/`). The production Sahyadri Config lacks `EVALPOTENTIAL`, so the production GADGET-4 snapshots most likely have no potential (not checked directly). Adding the flag left the ICs and output times byte-identical.

2. **consistent-trees needs many halos per snapshot**: it fails at 32^3 ("too few halos at scale factor 0.768278 to calculate consistency metric") even after relaxing `gen_merger_cfg.pl`'s 20000-byte catalogue threshold. At 256^3 the production threshold is used.

3. **VAHC is the slowest stage at 256^3**: 20–40 min per snapshot and process at grid 512^3 (FFT smoothing at many scales and per-halo loops), ∼23 GB per process. With 4 processes, 94 snapshots did not fit in 10 h; the jobs were rerun with 8 processes and 30 h (restartable: finished snapshots are skipped). At 2048^3 with 100 outputs this stage needs planning.

4. **Column names**: consistent-trees hlists call the columns `Mvir`, `Rvir`; Rockstar lists `ID`, `X`, ... The comparison scripts read both (fixed during the 256^3 run; the first two comparison jobs stopped on this).

5. **pkdgrav3 has no physical units when it reads an IC file** (`units.cxx`): box size in the output headers and `.pkc` potential units are wrong unless `dKpcUnit` and `dMsolUnit` are given; the pipeline's parameter generator now writes them. A Gadget-2 input file's IDs become iOrder directly, so they must be $0 \ldots N-1$; PKDGRAV3's HDF5 `Header/Time` is code time, not a. Recorded in the PKDGRAV3 fork, `docs/compressed_io_plan.md` section 7.

6. **FITS stage on large tree files** (VAHC rerun with `.vpot`): `compress_halo_outputs.py` verified float columns by comparing a sum over 2×10^6-row chunks with a single sum over the column, which differ in the last bits for the 2.5-million-row tree file ("column scale does not read back identically"). It then wrote `locations.fits.gz` with every `FitsRow` $= -1$, which a restart skipped. Both are fixed (chunked read-back sums; `FitsRow` from existing tree FITS files, and no `locations.fits.gz` while a tree lacks a row). Final check: 661 PASS, 0 FAIL per code. On pawna `qalter` cannot change a queued job's walltime, and the shared node made the VAHC rerun take ∼3.5 h for 94 snapshots with 4 processes.

7 Reproducing

```
# on pawna, in a clone of sahyadri-sandbox (branch rockstar-pkc-reader)
cd scripts/gadget_pkdgrav3_compare
bash -lc "bash setup_gadget4_pawna.sh <gadget4 source> <G4HOME> 256"   # Gadget-4 (once)
bash submit_compare.sh compare256.conf                                 # 9 PBS jobs
# figures (Mac or anywhere with matplotlib), from reports/gadget_pkdgrav3_potential:
rsync pawna:<data>/compare/pkd256_vs_gadget256_r1/{particles.csv,profiles,halos} \
    data/pkd256_vs_gadget256_r1/
jupyter nbconvert --to notebook --execute stack_potential.ipynb
python make_figures.py 32=data/pkd32_vs_gadget32_r1 256=<copy with snap_*.npz>
# .vpot: copies of <halos>/compressed/out_*_{basic,vahc}.fits.gz of both runs
D=data/pkd256_vs_gadget256_r1
python ../../scripts/gadget_pkdgrav3_compare/compare_vpot.py <gadget copy> <pkd copy> \
    $D/vpot/vpot_match.csv --particles $D/particles.csv
python make_vpot_figures.py
# halo-halo potential cross-correlation, z = 0 (on pawna: submit_pipeline.sh <conf> --stages xcorr)
for c in gadget256 pkd256; do for b in log:0.2,50,30 lin:0,50,0.2; do
  python ../../scripts/post-process/halo_potential_xcorr.py <halos>/$c/r1/compressed 100 \
      xc/${c}_out100_xc_${b%%:*}.npz --rbins $b; done; done
python make_xcorr_figures.py --full xc
# correlation multipoles (Corrfunc fork; on pawna: submit_pipeline.sh <conf> --stages ximult)
```

```
for c in gadget256 pkd256; do python ../../scripts/post-process/halo_xi_multipoles.py \
    <halos>/$c/r1/compressed 100 xm/${c}_out100_ximult.npz --los x,y,z; done
python ../../scripts/post-process/halo_xi_multipoles.py <halos>/pkd256/r1/compressed 100 \
    xm/pkd256_out100_amp50.npz --pot-amp 50 --spaces real,grav --pairs 0-1,0-2,0-3 --los x,y,z
python make_ximult_figures.py --full xm
latexmk -pdf report.tex
```